

163. Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4

163.1 Overview

This clause specifies the 100GBASE-KR1 PMD, the 200GBASE-KR2 PMD, the 400GBASE-KR4 PMD, and the baseband medium. The specifications for the three PMDs are similar, except for the number of lanes and associated parameters and the MDI.

There are two associated annexes. Annex 163A provides measurement methods and test points for backplane and chip-to-chip interfaces. Annex 163B provides information on an example test fixture.

When forming a complete Physical Layer, a PMD shall be connected as illustrated in Figure 163–1, to the appropriate sublayers (as specified in Table 163–1, Table 163–2, and Table 163–3), to the medium through the appropriate MDI, and to the management functions that are optionally accessible through the management interface defined in Clause 45, or equivalent.

Table 163–1—Physical Layer clauses associated with the 100GBASE-KR1 PMD

Associated clause	100GBASE-KR1
81—RS	Required
81—CGMII ^a	Optional
82—PCS for 100GBASE-R	Required
91—RS-FEC	Required
152—Inverse RS-FEC	Optional ^b
161—RS-FEC-Int	Required
83—PMA for 100GBASE-R	Optional ^c
83A—CAUI-10	Optional
83D—CAUI-4 C2C	Optional
135—PMA for 100GBASE-P	Required
135D—100GAUI-4 C2C	Optional
135F—100GAUI-2 C2C	Optional
120F—100GAUI-1 C2C	Optional
73—Auto-Negotiation	Required

^a The CGMII is an optional interface. However, if the CGMII is not implemented, a conforming implementation behaves functionally as though the RS and CGMII were present.

^b Inverse RS-FEC is required to convert between RS-FEC and RS-FEC-Int (see 152.1.2).

^c PMA for 100GBASE-R is required when either CAUI-10 or CAUI-4 is used.

Table 163–2—Physical Layer clauses associated with the 200GBASE-KR2 PMD

Associated clause	200GBASE-KR2
117—RS	Required
117—200GMII ^a	Optional
118—200GMII extender	Optional
119—PCS for 200GBASE-R	Required
120—PMA for 200GBASE-R	Required
120B—200GAUI-8 C2C	Optional
120D—200GAUI-4 C2C	Optional
120F—200GAUI-2 C2C	Optional
73—Auto-Negotiation	Required

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

Table 163–3—Physical Layer clauses associated with the 400GBASE-KR4 PMD

Associated clause	400GBASE-KR4
117—RS	Required
117—400GMII ^a	Optional
118—400GMII extender	Optional
119—PCS for 400GBASE-R	Required
120—PMA for 400GBASE-R	Required
120B—400GAUI-16 C2C	Optional
120D—400GAUI-8 C2C	Optional
120F—400GAUI-4 C2C	Optional
73—Auto-Negotiation	Required

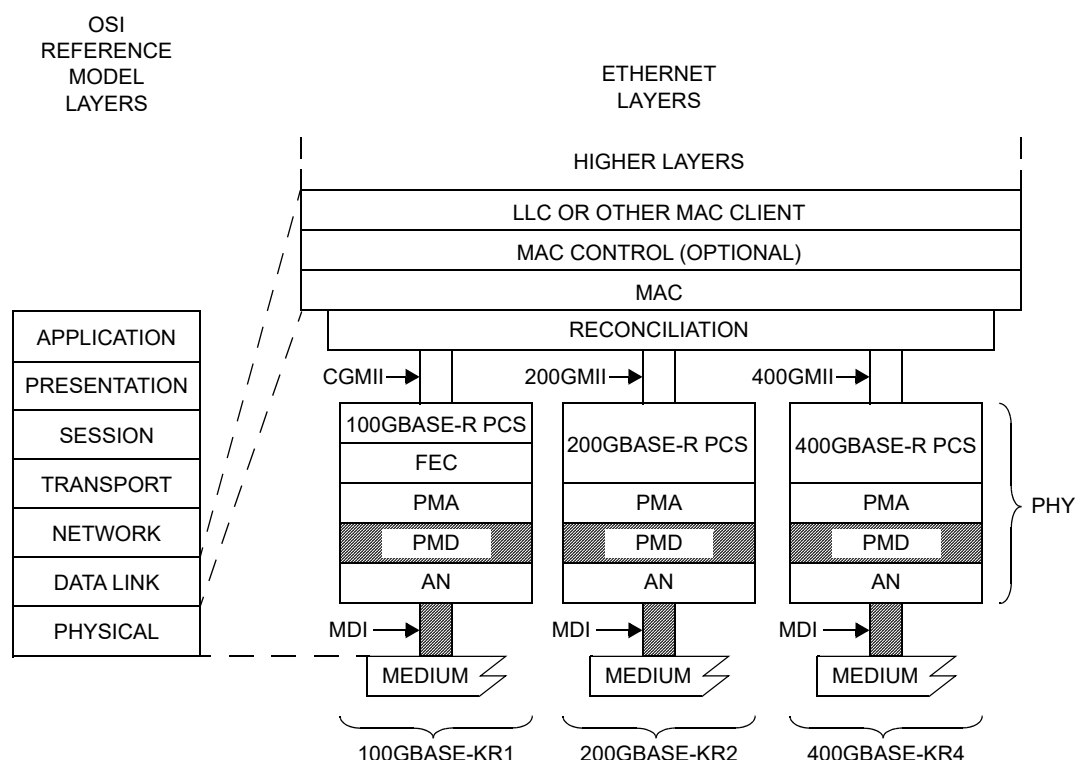
^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

For the 100GBASE-KR1 PHY, in order to support the required frame loss ratio (see 1.4.344) of less than 6.2×10^{-10} for 64-octet frames with minimum interpacket gap, the PMD and the adjacent PMA are expected to detect bits from a compliant input signal at a BER lower than 2.4×10^{-4} assuming errors are sufficiently uncorrelated. This BER allocation enables a frame loss ratio lower than 10^{-10} after processing by the FEC (Clause 91 or Clause 161) and the PCS (Clause 82) if there are negligible errors due to other electrical interfaces (100GAUI-n). If the PMD and PMA create errors that are not sufficiently uncorrelated, the BER is required to be lower as appropriate to maintain a frame loss ratio lower than 10^{-10} .

For the 200GBASE-KR2 and 400GBASE-KR4 PHYs, in order to support the required frame loss ratio (see 1.4.344) of less than 6.2×10^{-11} for 64-octet frames with minimum interpacket gap, the PMD and the adjacent PMA are expected to detect bits from a compliant input signal at a BER lower than 2.4×10^{-4} assuming errors are sufficiently uncorrelated. This BER allocation enables a frame loss ratio lower than 9.2×10^{-13} after processing by the PCS (Clause 119) if there are negligible errors due to other electrical interfaces (200GAUI-n or 400GAUI-n). If the PMD and PMA create errors that are not sufficiently uncorrelated, the BER is required to be lower as appropriate to maintain a frame loss ratio lower than 9.2×10^{-13} .

A compliant input signal is a transmitter output of a compliant PHY that has passed through a compliant channel.

Figure 163–1 shows the relationship of the PMD and MDI to the ISO/IEC Open System Interconnection (OSI) reference model.



200GMII = 200 Gb/s MEDIA INDEPENDENT INTERFACE
400GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE
AN = AUTO-NEGOTIATION
CGMII = 100 Gb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL

MDI = MEDIUM DEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 163–1—100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4 relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

163.2 Conventions

Clause 163 describes three PMDs, 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4, which have one, two, and four lanes, respectively. For efficient description, the parameter n is used to describe the number of lanes in a specific PMD. Accordingly, $n = 1$ for 100GBASE-KR1, $n = 2$ for 200GBASE-KR2, and $n = 4$ for 400GBASE-KR4.

The parameter i is used as an index or a suffix to identify a specific lane, and takes the values 0 to $n - 1$.

Within this clause, the unqualified term “PMD” refers to any of 100GBASE-KR1 PMD, 200GBASE-KR2 PMD, or 400GBASE-KR4 PMD.

Within this clause, the term FEC refers to either Clause 91 RS-FEC or the Clause 161 RS-FEC-Int for 100GBASE-KR1, or the RS-FEC within the Clause 119 PCS for 200GBASE-KR2 and 400GBASE-KR4.

163.3 PMD service interfaces

The service interfaces of the 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4 PMDs are identical to those of the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 PMDs, respectively (see 162.3).

163.4 PCS requirements for Auto-Negotiation (AN) service interface

The PCS associated with the PMD is required to support the AN service interface primitive AN_LINK.indication defined in 73.9. The requirements are the same as those specified in 162.4.

163.5 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the PMD and AN including the medium in one direction shall be no more than the maximum delays listed in Table 163–4. It is assumed that the one-way delay through the medium is no more than 14 ns.

Table 163–4—Delay constraints

PMD	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
100GBASE-KR1	4 096	8	40.96
200GBASE-KR2	8 192	16	40.96
400GBASE-KR4	16 384	32	40.96

^a One bit time is equal to 10 ps for 100GBASE-KR1, 5 ps for 200GBASE-KR2, and 2.5 ps for 400GBASE-KR4. (See 1.4.215 for the definition of bit time.)

^b One pause_quantum is equal to 5.12 ns for 100GBASE-KR1, 2.56 ns for 200GBASE-KR2, and 1.28 ns for 400GBASE-KR4. (See 31B.2 for the definition of pause_quanta.)

Descriptions of overall system delay constraints can be found in 80.4 for 100GBASE-KR1 and in 116.4 for 200GBASE-KR2 and 400GBASE-KR4.

163.6 Skew constraints

The Skew (relative delay) between the PCS or FEC lanes is kept within limits so that the information on the PCS or FEC lanes can be reassembled by the PCS or FEC. The Skew Variation is also limited to ensure that a given PCS or FEC lane always traverses the same physical lane.

163.6.1 Skew Constraints for 100GBASE-KR1

Skew and Skew Variation are defined in 80.5 and specified at the points SP1 to SP7 shown in [Figure 80–9](#).

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns as defined by [135.5.3.5](#). Since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns. Since the signal at the MDI represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns. Since the signal at the MDI represents a serial bit stream, there is no Skew Variation at this point.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns. Since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

For more information on Skew and Skew Variation, see 80.5. The measurements of Skew and Skew Variation are defined in [89.7.2](#).

163.6.2 Skew Constraints for 200GBASE-KR2 and 400GBASE-KR4

Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in [Figure 116–5](#).

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5.

163.7 PMD MDIO function mapping

The PMD MDIO function mapping requirements for the PMDs defined in this clause are identical to those of 162.7.

163.8 PMD functional specifications

163.8.1 Link block diagram

One direction of a 100GBASE-KR1, 200GBASE-KR2, or 400GBASE-KR4 link is shown in Figure 163–2. The PMDs on both ends of the link have connected ground references.

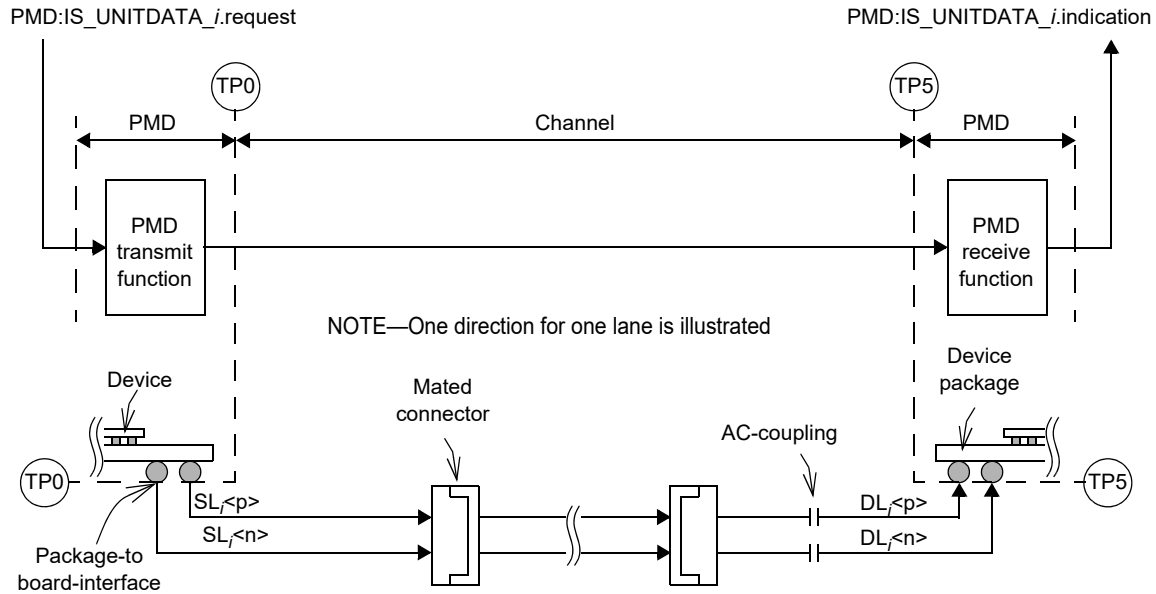


Figure 163–2—100GBASE-KR1, 200GBASE-KR2, or 400GBASE-KR4 link

163.8.2 PMD transmit function

The PMD transmit function specification is identical to that of 162.8.2 with the exception that electrical signals are delivered to the MDI, according to the transmit electrical specifications in 163.9.2.

163.8.3 PMD receive function

The PMD receive function specification is identical to that of 162.8.3 with the exception that electrical signals are received from the MDI, according to the receive electrical specifications in 163.9.3.

163.8.4 PMD global signal detect function

The PMD global signal detect function specification is identical to that of 162.8.4.

163.8.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function specification is identical to that of 162.8.5.

163.8.6 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional. Its specification is identical to that of 162.8.6.

163.8.7 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional. Its specification is identical to that of 162.8.7.

163.8.8 PMD fault function

The PMD fault function specification is identical to that of 162.8.8

163.8.9 PMD transmit fault function (optional)

The PMD transmit fault function is optional. Its specification is identical to that of 162.8.9.

163.8.10 PMD receive fault function (optional)

The PMD receive fault function is optional. Its specification is identical to that of 162.8.10.

163.8.11 PMD control function

The PMD control function specification is identical to that of 162.8.11.

163.9 Electrical characteristics

163.9.1 Reference impedance

The reference impedance for differential specifications is 100 Ω . The reference impedance for common-mode specifications is 25 Ω .

163.9.2 Transmitter characteristics

The transmitter shall meet the specifications given in Table 163–5. A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for all transmitter signal measurements, unless otherwise specified.

Table 163–5—Summary of transmitter specifications at TP0v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)		53.125 $\pm$ 50 ppm ^a	GBd
Differential pk-pk voltage (max) ^b Transmitter disabled Transmitter enabled	93.8.1.3	30 1200	mV mV
DC common-mode voltage (max) ^b	93.8.1.3	1	V
DC common-mode voltage (min) ^b	93.8.1.3	0.2	V
Low-frequency peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ (max)	162.9.4.4	30	mV
Signal to AC common-mode noise ratio, $SCMR$ (min)	163.9.2.6	15	dB
Difference effective return loss, $dERL$ (min)	163.9.2.2	–3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	163.9.2.3	3.25	dB
Difference steady-state voltage, dv_f (min)	163.9.2.4	0	V
Difference linear fit pulse peak ratio, dR_{peak} (min)	163.9.2.5	0	—
Level separation mismatch ratio, R_{LM} (min)	162.9.4.2	0.95	—

Table 163–5—Summary of transmitter specifications at TP0v (*continued*)

Parameter	Reference	Value	Units
Transmitter waveform			
absolute value of step size for all taps (min)	162.9.4.1.4	0.005	—
absolute value of step size for all taps (max)	162.9.4.1.4	0.025	—
value at minimum state for $c(-3)$ (max)	162.9.4.1.5	−0.06	—
value at maximum state for $c(-2)$ (min)	162.9.4.1.5	0.12	—
value at minimum state for $c(-1)$ (max)	162.9.4.1.5	−0.34	—
value at minimum state for $c(0)$ (max)	162.9.4.1.5	0.5	—
value at minimum state for $c(1)$ (max)	162.9.4.1.5	−0.2	—
Signal-to-noise-and-distortion ratio, SNDR (min)	162.9.4.6	32.5	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	162.9.4.3	28	dB
Jitter (max) ^c			
J_{RMS}	162.9.4.7	0.023	UI
J_{3u03}	162.9.4.7	0.106	UI
J_{3u}	162.9.4.7	0.115	UI
Even-odd jitter, pk-pk	162.9.4.7	0.025	UI

^a For a PMD in the same package as the PCS sublayer. In other cases, the signaling rate is derived from the input to the PMD transmit function provided by the adjacent PMA sublayer.

^b Measurement uses the method described in 93.8.1.3 with the exception that the PRBS13Q test pattern is used.

^c J_{3u} , J_{RMS} , and even-odd jitter measurements are made with a single transmit equalizer setting selected to compensate for the loss of the transmitter package and TP0 to TP0v test fixture.

163.9.2.1 Transmitter test fixture

Unless otherwise noted, measurements of the transmitter are made at the output of a test fixture (TP0v) as shown in Figure 163–3 and described in Annex 163A. An example test fixture is described in Annex 163B.

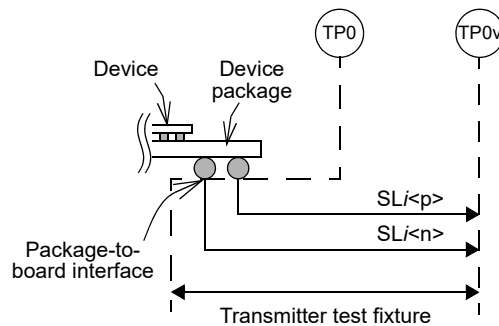


Figure 163–3—Transmitter test fixture and test points

163.9.2.1.1 Test fixture insertion loss

The insertion loss of the test fixture shall be between 1.7 dB and 5 dB at 26.56 GHz. The magnitude of the insertion loss deviation of the test fixture shall be less than or equal to 0.2 dB from 0.05 GHz to 26.56 GHz. Insertion loss deviation is calculated as specified in 93A.4, where T_i is 0.01 ns, and f_b and f_r values are taken from Table 163–11.

163.9.2.1.2 Test fixture effective return loss (ERL)

The ERL of the test fixture at TP0v is computed using the procedure in 93A.5 with the values in Table 163–6. Parameters that do not appear in Table 163–6 take values from Table 163–11.

Table 163–6—Test fixture ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	200	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

The ERL at TP0v shall be greater than or equal to 15 dB.

163.9.2.1.3 Test fixture common-mode to common-mode return loss

The common-mode to common-mode return loss of the test fixture shall be greater than or equal to 6 dB at all frequencies between 0.2 GHz and 40 GHz.

163.9.2.2 Transmitter difference ERL

The difference ERL of the transmitter at TP0v is computed using the procedure in 163A.3.2.2 with the values in Table 163–7 and Table 163–11, and with the value of T_{fx} equal to twice the delay from TP0 to TP0v.

The difference ERL at TP0v shall be greater than or equal to $dERL$ (min) in Table 163–5.

Table 163–7—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	200	UI
Equalizer length associated with reflection signal	N_{bx}	21	UI
Tukey window flag	tw	1	—

163.9.2.3 Transmitter common-mode to common-mode return loss

The common-mode to common-mode return loss shall be greater than or equal to RL_{cc} (min) in Table 163–5 at all frequencies between 0.2 GHz and 40 GHz.

163.9.2.4 Difference steady-state voltage

The difference steady-state voltage of the transmitter at TP0v is computed using the procedure in 163A.3.2.1 with $N_v = 200$ and other parameter values specified in Table 163–11.

The difference steady-state voltage at TP0v shall meet the specification dV_f (min) in Table 163–5.

163.9.2.5 Difference linear fit pulse peak ratio

The difference linear fit pulse peak ratio of the transmitter at TP0v is computed using the procedure in 163A.3.2.1 with parameters specified in Table 163–11.

The difference linear fit pulse peak ratio at TP0v shall meet the specification dR_{peak} (min) in Table 163–5.

163.9.2.6 Signal to AC common-mode noise ratio

Signal to AC common-mode noise ratio, $SCMR$, is calculated using Equation (163–1).

$$SCMR = 20\log_{10}\left(\frac{V_{peak}}{VCM_{FB}}\right) \quad (163-1)$$

where

$SCMR$	is the signal to AC common-mode noise ratio in dB
V_{peak}	is defined in 162.9.4.1.2
VCM_{FB}	is the full-band peak-to-peak AC common-mode voltage defined by the method specified in 162.9.4.4 and measured with the transmitter equalization set to “no equalization”

The signal to AC common-mode noise ratio shall meet the specification for $SCMR$ (min) in Table 163–5.

163.9.3 Receiver characteristics

Receiver electrical characteristics are specified at TP5v. The receiver shall meet the specifications given in Table 163–8.

Table 163–8—Summary of receiver specifications at TP5v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)	163.9.3.1	53.125 ± 100 ppm	GBd
Difference effective return loss, $dERL$ (min)	163.9.3.3	–3	dB
Differential-mode to common-mode return loss, RL_{cd}	163.9.3.4	Equation (163–2)	dB
Interference tolerance	163.9.3.5	Table 163–9	—
Jitter tolerance	163.9.3.6	Table 162–17	—

163.9.3.1 Receiver signaling rate

A PHY shall comply with the receiver requirements of 163.9.3.5 and 163.9.3.6 for any signaling rate in the range specified in Table 163–8.

NOTE—Although the PMD transmitter is specified with a signaling rate range of ± 50 ppm when in the same package as the PCS sublayer, the signaling rate range may be ± 100 ppm when derived from an intermediate interface (e.g., 100GAUI-4).

163.9.3.2 Receiver test fixture

Unless otherwise noted, measurements of the receiver are made at the input of a test fixture (TP5v) as shown in Figure 163–4.

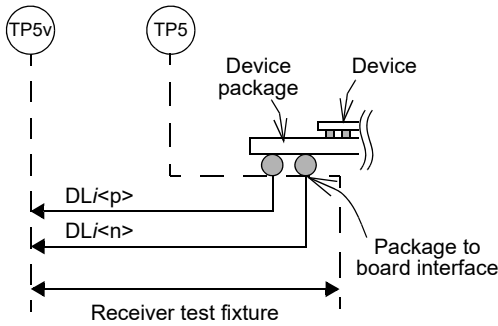


Figure 163–4—Receiver test fixture and test points

Receiver test fixture characteristics are the same as specified in 163.9.2.1.

163.9.3.3 Receiver difference ERL

The difference ERL of the receiver at TP5v is computed using the procedure in 163A.4.1.2 with the values in Table 163–7 and Table 163–11, and with the value of T_{fx} equal to twice the delay from TP5v to TP5.

The receiver difference ERL at TP5v shall be greater than or equal to $dERL$ (min) specified in Table 163–8.

163.9.3.4 Receiver differential-mode to common-mode return loss

The differential-mode to common-mode return loss of the receiver shall meet Equation (163–2). The differential-mode to common-mode return loss limit is illustrated by Figure 163–5.

$$RL_{cd}(f) \geq \begin{cases} 25 - 20(f/53.125) & 0.05 \leq f \leq 26.5625 \\ 15 & 26.5625 < f \leq 53.125 \end{cases} \quad (163-2)$$

where

$RL_{cd}(f)$ is the differential-mode to common-mode return loss at frequency f in dB
 f is the frequency in GHz

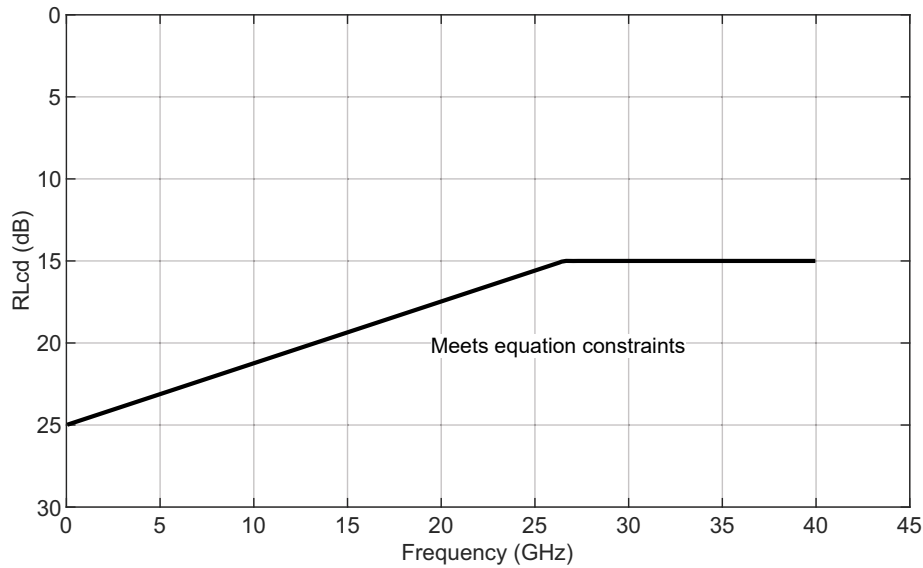


Figure 163-5—Receiver differential-mode to common-mode return loss limit

163.9.3.5 Receiver interference tolerance

Receiver interference tolerance is defined by the procedure in [Annex 93C](#). The receiver on each lane shall meet the FEC symbol error ratio requirement with channels matching the Channel Operating Margin (COM) and loss parameters for Test 1 and Test 2 in Table 163-9. The following additional considerations apply to the interference tolerance test.

- a) TP0v (TP5v) replaces TP0a (TP5a) in Annex 93A and [Annex 93C](#).
- b) The test transmitter is constrained such that for any transmitter equalizer setting the differential peak-to-peak voltage (see [93.8.1.3](#)) is less than or equal to 800 mV.
- c) The ERL of the test setup in [Figure 93C-4](#) measured at TP5 replica towards TPt meets the requirements in 163.10.3.
- d) The lower frequency bound for the noise spectral density constraints, f_{NSDI} , is 1 GHz.
- e) For the calculation of test channel COM, the transmitter model is determined in one of the following ways.
 - If the transmitter is a device with known S-parameters and transition time T_r , these parameters should be used instead of the transmitter package model in 93A.1.2. T_r should be provided as the value at the input of the device S-parameters network, as defined in 120G.3.1.4 but with no observation filter.
 - If a calibrated instrument-grade transmitter is used, the TP0 to TP0a trace in [Figure 93C-2](#) and [Figure 93C-3](#) and TP0 to TP0a replica trace in [Figure 93C-4](#) are omitted and the transmitter device package model $S^{(tp)}$ is omitted from [Equation \(93A-3\)](#). The filtered voltage transfer function $H^{(k)}(f)$ calculated in [Equation \(93A-19\)](#) uses the filter $H_f(f)$ defined by [Equation \(93A-46\)](#), where T_r is the measured transmitter transition time (see 120G.3.1.4).
 - If the transmitter is composed of a device with unknown S-parameters or unknown transition time, then the transmitter device package model $S^{(tp)}$ in 93A.1.2 is used, and T_r is determined from measurement at TP0v and the TP0 to TP0v S-parameters. The transmitter transition time (see 120G.3.1.4) is measured at TP0v with transmit equalization turned off by setting coefficients to preset 1 values (see 162.9.4.1.3). T_r is set as the value in [Equation \(93A-46\)](#) that

- would result in the reference transition time $T_r^{(ref)}$, determined according to 163A.3.1.3 with f_b and A_v equal to values in Table 163–11, being equal to the measured transition time.
- f) For the calculation of test channel COM, the following parameters are based on values measured from the test transmitter.
 - The parameter SNR_{TX} is set to the measured value of SNDR with $N_p = 29$.
 - The parameter R_{LM} is set to the measured value of R_{LM} .
 - The parameters A_{DD} and σ_{RJ} are calculated from the measured values of $J3u$ and J_{RMS} using Equation (163–3) and Equation (163–4) respectively.
 - g) Other COM parameters are set according to the values in Table 163–11.
 - h) COM is calculated using both Test 1 and Test 2 device package model transmission line lengths listed in Table 163–11 on the receiver side. The value of COM is taken as the lower of the two calculated values.
 - i) The test pattern is the scrambled idle test pattern defined in 119.2.4.9.
 - j) As an alternative to using the scrambled idle test pattern and measuring FEC symbol error ratio it is permissible to use the PRBS31Q pattern as described in 120.5.11.2.2 and bit error ratio testing. In this case the required bit error ratio is equal to the required FEC symbol error ratio divided by 10. Note that this requirement can be somewhat more stringent than using the scrambled idle test pattern and measuring FEC symbol error ratio, and therefore failing this test requirement with the PRBS31Q pattern does not necessarily imply a failure of the interference tolerance test.
 - k) A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for measurement of the signal applied by the pattern generator and for measurements of the broadband noise.

$$A_{DD} = \left(\frac{J3u}{2} + Q3d \sqrt{\left((Q3d^2 + 1) \times J_{RMS}^2 - \left(\frac{J3u}{2} \right)^2 \right)} \right) / (Q3d^2 + 1) \quad (163-3)$$

$$\sigma_{RJ} = \left(\frac{J3u}{2} - A_{DD} \right) / (Q3d) \quad (163-4)$$

where

$$Q3d = 3.0902$$

NOTE 1— $Q3d$ is an approximated solution of $Q(Q3d) = 1 \times 10^{-3}$, where the Q function is defined in Equation (95–1).

NOTE 2—Calculation of A_{DD} requires that $(Q3d^2 + 1) \times J_{RMS}^2 \geq \left(\frac{J3u}{2} \right)^2$. If this does not hold, a different transmitter should be used in the test setup.

Table 163–9—Receiver interference tolerance parameters

Parameter	Test 1 (low loss)			Test 2 (high loss)			Units
	Min	Max	Target	Min	Max	Target	
FEC symbol error ratio ^a	—	10^{-3}	—	—	10^{-3}	—	—
Insertion loss, IL_{dd} , at 26.5625 GHz ^b	13.5	14.5	—	27.5	28.5	—	dB
RSS_DFE4 ^c	0.05	—	—	0.05	—	—	—
COM including effects of broadband noise ^d	—	—	3	—	—	3	dB

^a The FEC symbol error ratio is measured in step 10) of the receiver interference tolerance method defined in 93C.2

^b Measured between TPt and TP5 (see Figure 93C–4).

^c RSS_DFE4 is described in 93A.2.

^d The COM value is the target for the receiver noise level calibration defined in 93C.2 step 7). The channel noise voltage applied in 93C.2 step 8) should be as close as practical to the value needed to produce the target COM. If higher amplitude values are used, this would demonstrate margin to the specification but this is not required for compliance.

163.9.3.6 Receiver jitter tolerance

Receiver jitter tolerance is verified for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 162–17. The test setup shown in Figure 93–12, or its equivalent, is used. TP0v (TP5v) replaces TP0a (TP5a) in Annex 93A, Annex 93C, and Annex 120D. The test channel meets the insertion loss requirement for Test 2 in Table 163–9. The synthesizer frequency is set to the specified jitter frequency and the synthesizer output amplitude is adjusted until the specified peak-to-peak jitter amplitude for that frequency is measured at TP0v. The test procedure is the same as the one described in 163.9.3.5, with the following exceptions:

- No broadband noise is added.
- The test channel COM, calculated per items 3) through 7) in 93C.2, is at least 3 dB.
- For the COM parameter calibration described in 93C.2 item 7), the test channel transmitter J_{RMS} and J_{3u} values are measured with the jitter frequency and amplitude set according to Case F from Table 162–17.

The receiver under test shall meet the FEC symbol error ratio in Table 163–9 for each case in Table 162–17.

163.10 Channel characteristics

Table 163–10 provides a summary of the channel characteristics for 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4, and references to the subclauses addressing each parameter.

163.10.1 Channel operating margin

The Channel Operating Margin (COM) is computed using the procedure in 93A.1 with the values in Table 163–11, where T_r is 7.5 ps for $H_t(f)$ as used in Equation (93A–19). COM is calculated twice, Test 1 and Test 2, with different values of device package model transmission line length z_p . COM shall be greater than or equal to 3 dB for both Test 1 and Test 2.

Table 163–10—Channel characteristics summary

Description	Reference	Value	Unit
Minimum COM	163.10.1	3	dB
Maximum insertion loss, IL_{dd} , at 26.56 GHz (recommended)	163.10.2	28	dB
Minimum channel ERL	163.10.3	9.7	dB
Differential-mode to common-mode return loss, RL_{cd}	163.10.4	Equation (163–6)	dB
Differential-mode to common-mode insertion loss, IL_{cd}	163.10.5	Equation (163–7)	dB
Common-mode to differential-mode insertion loss, IL_{dc}	163.10.6	Equation (163–8)	dB
Maximum AC-coupling 3 dB corner frequency	163.10.7	50	kHz

Table 163–11—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	53.125	GBd
Maximum start frequency	$f_{\min}$	0.05	GHz
Maximum frequency step	Δf	0.01	GHz
Device package model			
Single-ended device capacitance	C_d	1.2×10^{-4}	nF
Single-ended device series inductance	L_s	0.12	nH
Single-ended bump capacitance	C_b	3×10^{-5}	nF
Transmission line length, Test 1	z_p	12	mm
Transmission line length, Test 2 Tx	z_p	31	mm
Transmission line length, Test 2 Rx	z_p	29	mm
Transmission line parameter, a_1	a_1	9.909×10^{-4}	ns ^{1/2} /mm
Transmission line parameter, a_2	a_2	2.772×10^{-4}	ns/mm
Single-ended package capacitance at package-to-board interface	C_p	8.7×10^{-5}	nF
Transmission line characteristic impedance	Z_c	87.5	Ω
Transmission line 2 length	z_{p2}	1.8	mm
Transmission line 2 characteristic impedance	Z_{c2}	92.5	Ω
Single-ended reference resistance	R_0	50	Ω
Single-ended termination resistance	R_d	50	Ω
Receiver 3 dB bandwidth	f_r	$0.75 \times f_b$	GHz
Transmitter equalizer, minimum cursor coefficient	$c(0)$	0.54	—
Transmitter equalizer, 3 rd pre-cursor coefficient	$c(-3)$	—	—
Minimum value		–0.06	
Maximum value		0	
Step size		0.02	
Transmitter equalizer, 2 nd pre-cursor coefficient	$c(-2)$	—	—
Minimum value		0	
Maximum value		0.12	
Step size		0.02	

Table 163–11—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Transmitter equalizer, 1 st pre-cursor coefficient Minimum value Maximum value Step size	$c(-1)$	–0.34 0 0.02	—
Transmitter equalizer, post-cursor coefficient Minimum value Maximum value Step size	$c(1)$	–0.2 0 0.02	—
Continuous time filter, DC gain Minimum value Maximum value Step size	g_{DC}	–20 0 1	dB dB dB
Continuous time filter, DC gain 2 Minimum value Maximum value Step size	g_{DC2}	–6 0 1	dB dB dB
Continuous time filter, zero frequency for $g_{DC} = 0$	f_z	$f_b / 2.5$	GHz
Continuous time filter, pole frequencies	f_{p1} f_{p2}	$f_b / 2.5$ f_b	GHz GHz
Continuous time filter, low-frequency pole/zero	f_{LF}	$f_b / 80$	GHz
Transmitter differential peak output voltage Victim Far-end aggressor Near-end aggressor	A_v A_{fe} A_{ne}	0.413 0.413 0.608	V V V
Number of signal levels	L	4	—
Level separation mismatch ratio	R_{LM}	0.95	—
Transmitter signal-to-noise ratio	SNR_{TX}	33	dB
Number of samples per unit interval	M	32	—
Decision feedback equalizer (DFE) length	N_b	12	UI
Normalized DFE coefficient maximum limit $n = 1$ $n = 2$ $n = 3$ to 6 $n = 7$ to N_b	$bb_{\max}(n)$	0.85 0.3 0.2 0.1	—
Normalized DFE coefficient minimum limit $n = 1$ $n = 2$ $n = 3$ to N_b	$bb_{\min}(n)$	0.3 0.05 –0.03	—
Number of DFE floating tap banks	N_{bg}	3	—
Number of DFE floating taps per bank	N_{bf}	3	—
DFE maximum span including floating taps	N_f	40	UI
Normalized coefficient magnitude limit for DFE floating taps	$b_{g\max}$	0.05	—
DFE floating tap tail root-sum-of-squares limit	$\sigma_{t\max}$	0.02	—

Table 163–11—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
DFE floating tap tail starting position	N_{ts}	25	—
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI
One-sided noise spectral density	η_0	8.2×10^{-9}	V ² /GHz
Target detector error ratio	DER_0	10^{-4}	—

163.10.2 Channel insertion loss (recommended)

The channel insertion loss should meet Equation (163–5).

$$ILdd(f) \leq \begin{cases} 0.693 + 2.161\sqrt{f} + 0.607f & 0.01 \leq f \leq 26.5625 \\ -19.12 + 1.773f & 26.5625 < f \leq 40 \end{cases} \quad (163-5)$$

where

$ILdd(f)$ is the insertion loss in dB at frequency f
 f is the frequency in GHz

The insertion loss limit is illustrated by Figure 163–6.

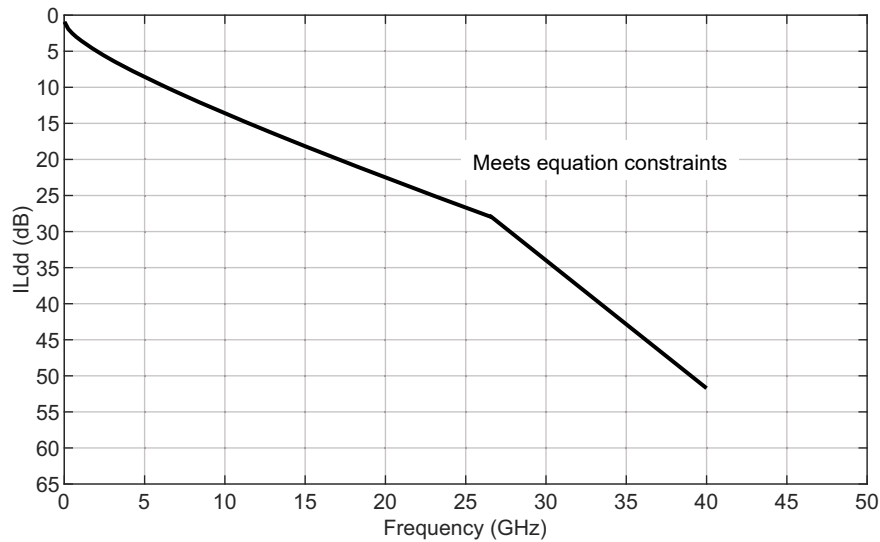


Figure 163–6—Channel insertion loss limit

163.10.3 Channel ERL

The ERL of the channel at TP0 and at TP5 is computed using the procedure in 93A.5 with the values in Table 163–12 and Table 163–11.

Channel ERL at TP0 and at TP5 shall be greater than or equal to 9.7 dB.

Table 163–12—Channel ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	3500	UI
Equalizer length associated with reflection signal	N_{bx}	21	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

163.10.4 Channel differential-mode to common-mode return loss

The TP0 to TP5 channel differential-mode to common-mode return loss shall meet Equation (163–6).

$$RLcd(f) \geq \begin{cases} 22 - 10(f/26.56) & 0.05 \leq f < 26.56 \\ 15 - 3(f/26.56) & 26.56 \leq f \leq 40 \end{cases} \quad (163-6)$$

where

$RLcd(f)$ is the TP0 to TP5 channel differential-mode to common-mode return loss at frequency f in dB

f is the frequency in GHz

The TP0 to TP5 channel differential-mode to common-mode return loss limit is illustrated in Figure 163–7.

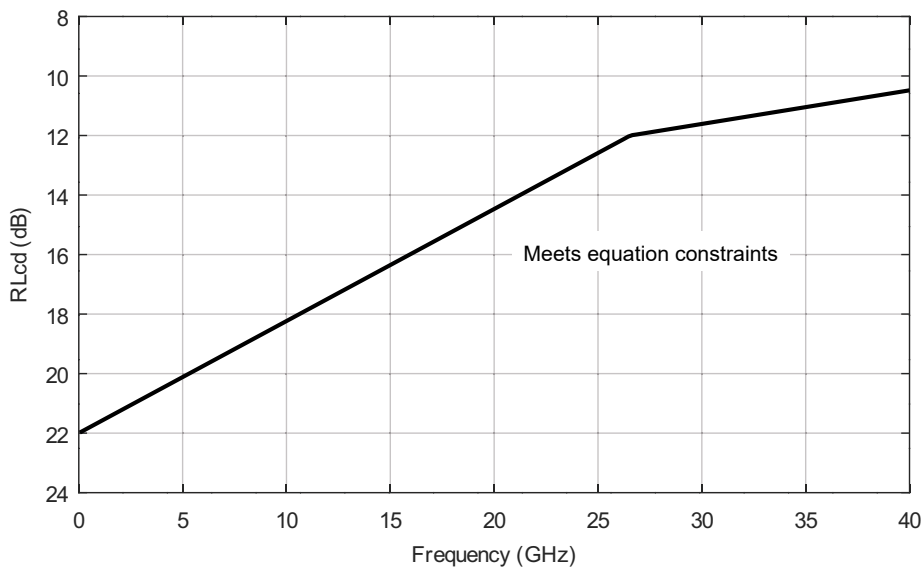


Figure 163-7— Channel differential-mode to common-mode return loss

163.10.5 Channel differential-mode to common-mode insertion loss

The difference between the TP0 to TP5 channel differential-mode to common-mode insertion loss and the TP0 to TP5 channel insertion loss shall meet Equation (163-7) as illustrated in Figure 163-8.

$$ILcd(f) - ILdd(f) \geq \left\{ \begin{array}{ll} 10 & 0.05 \leq f < 12.89 \\ 14 - 0.3108f & 12.89 \leq f \leq 40 \end{array} \right\} \quad (163-7)$$

where

- $ILcd(f)$ is the TP0 to TP5 channel differential-mode to common-mode insertion loss at frequency f in dB
- $ILdd(f)$ is the TP0 to TP5 insertion loss at frequency f in dB
- f is the frequency in GHz

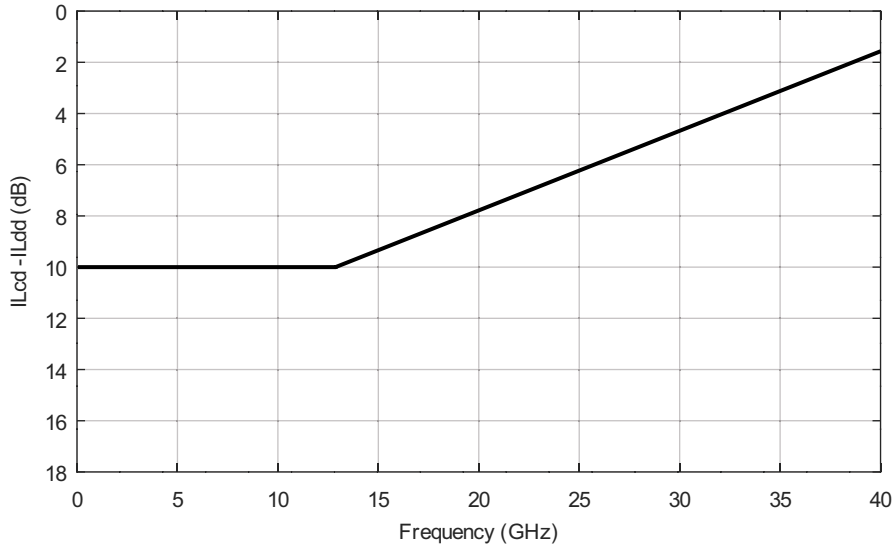


Figure 163–8—Channel differential-mode to common-mode insertion loss to insertion loss difference

163.10.6 Channel common-mode to differential-mode insertion loss

The difference between the TP0 to TP5 channel common-mode to differential-mode insertion loss and the TP0 to TP5 channel insertion loss shall meet Equation (163–8) illustrated in Figure 163–9.

$$ILdc(f) - ILdd(f) \geq \begin{cases} 10 & 0.05 \leq f < 12.89 \\ 14 - 0.3108f & 12.89 \leq f \leq 40 \end{cases} \quad (163-8)$$

where

- $ILdc(f)$ is the TP0 to TP5 channel common-mode to differential-mode insertion loss at frequency f in dB
- $ILdd(f)$ is the TP0 to TP5 insertion loss at frequency f in dB
- f is the frequency in GHz

163.10.7 AC-coupling

AC-coupling shall be implemented within the channel (between TP0 and TP5) using DC blocking capacitors. The low-frequency 3 dB cutoff of the channel shall be less than 50 kHz.

Systems with no AC-coupling within the channel are considered engineered links. It is the system integrator’s responsibility to verify that the transmitter and the receiver are compatible with the common-mode voltage differences that may exist in this configuration.

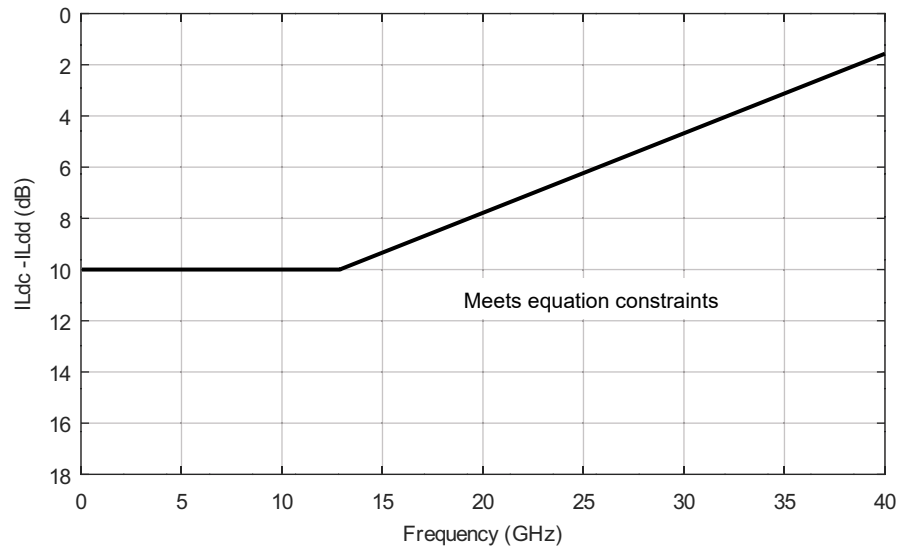


Figure 163-9—Channel common-mode to differential-mode insertion loss to insertion loss difference

163.11 MDI specifications

The MDI for the 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4 PHYs is an implementation-dependent direct electrical connection between the PMD and the medium. The MDI is composed of $2 \times n$ differential pairs, one pair for the transmit function and one pair for the receive function on each lane, marked by TP0 and TP5 in Figure 163-2.

163.12 Environmental specifications

All equipment subject to this clause shall conform to the applicable requirements of [93.10](#).

163.13 Protocol implementation conformance statement (PICS) proforma for Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4¹⁹

163.13.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

163.13.2 Identification

163.13.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

163.13.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
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¹⁹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

163.13.3 Major capabilities/options

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*KR1	100GBASE-KR1 PMD	163.1	Can operate as a 100GBASE-KR1 PMD	O.1	Yes [] No []
*KR2	200GBASE-KR2 PMD	163.1	Can operate as a 200GBASE-KR2 PMD	O.1	Yes [] No []
*KR4	400GBASE-KR4 PMD	163.1	Can operate as a 400GBASE-KR4 PMD	O.1	Yes [] No []
PMA100	100GBASE-R PMA	163.1	Device implements Clause 135 PMA for 100GBASE-P	KR1:M	Yes [] N/A []
PMA200	200GBASE-P PMA	163.1	Device implements Clause 120 PMA for 200GBASE-R	KR2:M	Yes [] N/A []
PMA400	400GBASE-R PMA	163.1	Device implements Clause 120 PMA for 400GBASE-R	KR4:M	Yes [] N/A []
*AUIPMD	100GAUI-n C2C, 200GAUI-n C2C, or 400GAUI-n C2C	163.1	Service interface of PMA adjacent to PMD is physically instantiated	O.2	Yes [] No []
AUIFEC	CAUI-n C2C or 100GAUI-n C2C	163.1	Service interface of PMA between PCS and RS-FEC is physically instantiated	!KR4:O.2	Yes [] No [] N/A []
*nGMII	CGMII, 200GMII, or 400GMII and RS	163.1	Interface is implemented or functionally equivalent	O.2	Yes [] No []
AN	Auto-Negotiation	163.1	Device implements Auto-Negotiation	M	Yes []
FEC100	100GBASE-R RS-FEC 100GBASE-P RS-FEC	163.1	Device implements Clause 91 and Clause 161 RS-FEC with RS(544,514)	KR1* !AUIPMD :M	Yes [] N/A []
PCS100	100GBASE-R PCS	163.1	Device implements Clause 82 100GBASE-R PCS	KR1* nGMII:M	Yes [] N/A []
PCS200	200GBASE-R PCS	163.1	Device implements Clause 119 200GBASE-R PCS	KR2* nGMII:M	Yes [] N/A []
PCS400	400GBASE-R PCS	163.1	Device implements Clause 119 400GBASE-R PCS	KR4* nGMII:M	Yes [] N/A []
PCSAN	PCS requirements for AN service interface	163.4	PCS supports service interface primitive AN_LINK.indication	nGMII:M	Yes [] N/A []
DC	Delay constraints	163.5	Conforms to constraints	M	Yes []
SC	Skew and Skew Variation constraints	163.6	Conforms to constraints	M	Yes []
*MD	MDIO capability	163.7	Device implements Clause 45 MDIO	O	Yes [] No []

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*CHNL	Channel	163.10	Item marked with CHNL are channel specifications, these items are the only ones applicable to channels and are not applicable to PHYs.	O	Yes [] No []
*PCSPKG	PMD in same package as PCS	163.9.2		O	Yes [] No []

^a A “*” preceding an “Item” identifier indicates there are other PICS that depend on whether or not this item is supported.

163.13.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4

163.13.4.1 Functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
PF1	Transmit function in DATA mode	162.8.2	Converts each symbol stream from the PMD service interface into an electrical signal as specified, and delivers it to the MDI, with the electrical specifications in 163.9.2	M	Yes []
PF2	Transmit function in TRAINING mode	162.8.2	Converts each symbol stream generated by the PMD control function into an electrical signal as specified, and delivers it to the MDI, with the electrical specifications in 163.9.2	M	Yes []
PF4	Receive function	162.8.3	Converts each electrical signal from the MDI into a symbol stream as specified, and delivers it to the PMD service interface	M	Yes []
PF5	Lane-by-lane signal detect	162.8.5	Set to one if training is disabled, otherwise set according to signal_detect	M	Yes []
PF6	Global PMD transmit disable	162.8.6	Function is implemented as specified	O	Yes [] No []
PF7	Lane-by-lane PMD transmit disable	162.8.7	Function is implemented as specified	O	Yes [] No []
PF8	PMD fault	162.8.8	PMD_fault variable mapped to the fault bit as specified in 45.2.1.2.3	MD:M	Yes [] N/A []
PF9	PMD transmit fault	162.8.9	Implemented, PMD_transmit_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.4	MD:O	Yes [] No [] N/A []
PF10	PMD receive fault	162.8.10	Implemented, PMD_receive_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.5	MD:O	Yes [] No [] N/A []

163.13.4.2 PMD control function

Item	Feature	Subclause	Value/Comment	Status	Support
PC1	PMD control function	162.8.11	Implemented as specified, one instance for each lane, operating independently	M	Yes []
PC2	Training pattern	136.8.11.1.3	Each lane implements four generator polynomials defined in Table 136–8	M	Yes []
PC3	Training pattern	136.8.11.1.3	State set to the value of seed <i>i</i> at the start of the training pattern	M	Yes []
PC4	Control field structure	162.8.11	As shown in 162.8.11	M	Yes []
PC5	Receiver frame lock bit	136.8.11.3.3	Initially set to zero, not set to 1 until local_tf_lock is true	M	Yes []
PC6	Initial condition setting	136.8.11.4.1	When requested, set according to the request, with values per Table 162–12	M	Yes []
PC7	Handshake timing	136.8.11.6	When the transmitted frame lock bit is 1, acknowledge requests within less than 2 ms	M	Yes []
PC8	Transmit precoded data	136.8.11.7.5	PMD causes adjacent PMA to use or not use precoding on transmitted data according to modulation and precoding status bit	M	Yes []
PC9	Receive precoded data	136.8.11.7.5	PMD informs adjacent PMA about precoding of received data according to modulation and precoding request bit	M	Yes []
PC10	Training	136.8.11.7.1	The value of use_quiet_in_training is TRUE.	M	Yes []

163.13.4.3 Transmitter characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Signaling rate, each lane, PMD in same package as PCS	163.9.2	Per Table 163–5	PCSP KG:M	Yes [] N/A []
TC2	Signaling rate, each lane, PMD not in same package as PCS	163.9.2	Derived from input to PMD transmit function	!PCSP KG:M	Yes [] N/A []
TC3	TP0 to TP0v test fixture	163.9.2.1	Meets specifications	M	Yes []
TC4	Output voltage specifications	93.8.1.3	Per Table 163–5	M	Yes []
TC5	Difference ERL at TP0v	163.9.2.2	Per Table 163–5	M	Yes []
TC6	Common-mode to common-mode return loss	93.8.1.4	Per Table 163–5	M	Yes []
TC7	Difference steady-state voltage	163.9.2.4	Per Table 163–5	M	Yes []
TC8	Difference linear fit pulse peak ratio	163.9.2.5	Per Table 163–5	M	Yes []
TC9	Level separation mismatch ratio	163.9.2	Per Table 163–5	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TC10	Absolute coefficient step size	162.9.4.1.4	Per Table 163–5	M	Yes []
TC11	Coefficient range	162.9.4.1.5	Per Table 163–5	M	Yes []
TC12	Signal-to-noise-and-distortion ratio	163.9.2	Per Table 163–5	M	Yes []
TC13	Output jitter	162.9.4.7	Per Table 163–5	M	Yes []
TC14	Transmitter signaling rate	163.9.2	Per Table 163–5	M	Yes []
TC15	Low-frequency peak-to-peak AC common-mode voltage	163.9.2	Per Table 163–5	M	Yes []
TC16	Signal to AC common-mode noise ratio	163.9.2	Per Table 163–5	M	Yes []
TC17	Signal-to-residual-intersymbol-interference ratio	163.9.2	Per Table 163–5	M	Yes []

163.13.4.4 Receiver characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	TP5 to TP5v test fixture	163.9.3.2	Meets specifications	M	Yes []
RC2	Difference ERL at TP5v	163.9.3.3	Greater than or equal to –3 dB	M	Yes []
RC3	Input differential-mode to common-mode return loss	163.9.3.4	Meets constraints of Equation (163–2)	M	Yes []
RC4	Interference tolerance	163.9.3.5	PHY meets FEC symbol error ratio requirement in all tests defined in Table 163–9	M	Yes []
RC5	Jitter tolerance	163.9.3.6	PHY meets FEC symbol error ratio requirement in all cases defined in Table 162–17	M	Yes []

163.13.4.5 Channel characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
CC1	ERL	163.10.3	Greater than or equal to 9.7 dB	CHNL:M	Yes [] N/A []
CC2	Channel Operating Margin (COM)	163.10	Greater than or equal to 3 dB	CHNL:M	Yes [] N/A []
CC3	AC-coupling	93.9.4	Between TP0 and TP5, 3 dB cutoff frequency less than 50 kHz	CHNL:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
CC4	Channel differential-mode to common-mode return loss	163.10.4	Per Equation (163–6)	CHNL:M	Yes [] N/A []
CC5	Channel differential-mode to common-mode insertion loss	163.10.5	Per Equation (163–7)	CHNL:M	Yes [] N/A []
CC6	Channel common-mode to differential-mode insertion loss	163.10.6	Per Equation (163–8)	CHNL:M	Yes [] N/A []

163.13.4.6 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	Environmental specifications	163.12	Conform to applicable requirements of 93.10	M	Yes []

Clause 167 was added to IEEE Std 802.3-2022 by IEEE Std 802.3db-2022.

167. Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4

167.1 Overview

Change Table 167–1 and Table 167–2 as follows (some unchanged rows not shown):

Table 167–1—Physical Layer clauses associated with the 100GBASE-VR1 and 100GBASE-SR1 PMDs

Associated clause	100GBASE-VR1, 100GBASE-SR1
...	
83E—CAUI-4 C2M	Optional
120F—100GAUI-1 C2C	<u>Optional</u>
120G—100GAUI-1 C2M	<u>Optional</u>
135—PMA for 100GBASE-P	Required
...	

Table 167–2—Physical Layer clauses associated with the 200GBASE-VR2, 400GBASE-VR4, 200GBASE-SR2, and 400GBASE-SR4 PMDs

Associated clause	200GBASE-VR2, 200GBASE-SR2	400GBASE-VR4, 400GBASE-SR4	Status
...			
120E	200GAUI-4 C2M	400GAUI-8 C2M	Optional
120F	<u>200GAUI-2 C2C</u>	<u>400GAUI-4 C2C</u>	<u>Optional</u>
120G	<u>200GAUI-2 C2M</u>	<u>400GAUI-4 C2M</u>	<u>Optional</u>
78	Energy-Efficient Ethernet		Optional

Annex 93A

(normative)

Specification methods for electrical channels

93A.1 Channel Operating Margin

Change Table 93A–1 as follows (some unchanged rows not shown):

Table 93A–1—COM parameters

Parameter	Reference	Symbol	Units
Signaling rate	93A.1.1	f_b	GBd
Maximum start frequency	93A.1.1	$f_{\min}$	GHz
Maximum frequency step	93A.1.1	Δf	GHz
Device package model Single-ended device capacitance ^a <u>Single-ended device series inductance^a</u> <u>Single-ended bump capacitance^a</u> Transmission line length ^a Single-ended package capacitance at package-to-board interface ^a Transmission line characteristic impedance ^a <u>Transmission line 2 length^a</u> <u>Transmission line parameter, a_1^b</u> <u>Transmission line parameter, a_2^b</u> <u>Transmission line 2 characteristic impedance^a</u>	93A.1.2	C_d L_s C_b z_p C_p Z_c z_{p2} a_1 a_2 Z_{c2}	nF nH nF mm nF Ω mm ns ^{1/2} /mm ns/mm Ω
Single-ended reference resistance	93A.1.2	R_0	Ω
Single-ended termination resistance	93A.1.3	R_d	Ω
Receiver 3 dB bandwidth	93A.1.4.1	f_r	GHz
Transmitter equalizer, minimum cursor coefficient	93A.1.4.2	$c(0)$	—
<u>Transmitter equalizer, 3rd pre-cursor coefficient^c</u> <u>Minimum value</u> <u>Maximum value</u> <u>Step size</u>	93A.1.4.2	<u>$c(-3)$</u>	<u>—</u> <u>—</u> <u>—</u>
Transmitter equalizer, 2 nd pre-cursor coefficient ^c Minimum value Maximum value Step size	93A.1.4.2	$c(-2)$	— — —
Transmitter equalizer, pre-cursor coefficient Minimum value Maximum value Step size	93A.1.4.2	$c(-1)$	— — —

Table 93A-1—COM parameters (*continued*)

Parameter	Reference	Symbol	Units
Transmitter equalizer, post-cursor coefficient	93A.1.4.2	$c(1)$	—
Minimum value			—
Maximum value			—
Step size			—
...			
Decision feedback equalizer (DFE) length	93A.1.6	N_b	UI
Normalized DFE coefficient magnitude limit ^d	93A.1.6	$b_{\max}(n)$	—
Normalized DFE coefficient maximum limit ^d	93A.1.6	$bb_{\max}(n)$	=
Normalized DFE coefficient minimum limit ^d	93A.1.6	$bb_{\min}(n)$	=
Number of DFE floating tap banks ^d	93A.1.6	N_{bg}	=
Number of DFE floating taps per bank ^d	93A.1.6	N_{bf}	=
DFE maximum span including floating taps ^d	93A.1.6	N_f	UI
Normalized coefficient magnitude limit for DFE floating taps ^d	93A.1.6	$b_{g\max}$	=
DFE floating tap tail root-sum-of-squares limit ^d	93A.1.6.1	σ_{imax}	=
DFE floating tap tail starting position ^d	93A.1.6.1	N_{ts}	=
Random jitter, RMS	93A.1.6	σ_{RJ}	UI
...			

^aSome clauses that invoke this method do not provide a value for C_d , L_s , C_{b1} , z_{p1} , C_{p2} , Z_{c2} , z_{p2} , or Z_{c2} . See 93A.1.2.

^bSome clauses that invoke this method do not provide a value for a_1 or a_2 . See 93A.1.2.3.

^cSome clauses that invoke this method do not provide a value for $c(-2)$ or $c(-3)$. See 93A.1.6.

^dSome clauses that invoke this method do not provide a value for $b_{\max}(n)$, $bb_{\max}(n)$, $bb_{\min}(n)$, N_{bg} , N_{bf} , N_f , $b_{g\max}$, σ_{imax} , or N_{ts} . See 93A.1.6.

Change Table 93A–2 as follows (some unmodified rows are not shown):

Table 93A–2—Physical Layer specifications that employ COM

Physical Layer	Parameter values
...	
100GBASE-CR2 (Clause 136)	Table 136–18
100GBASE-CR1 (Clause 162)	Table 162–20
100GBASE-KR4 (Clause 93)	Table 93–8
100GBASE-KP4 (Clause 94)	Table 94–17
100GBASE-KR2 (Clause 137)	Table 137–6
100GBASE-KR1 (Clause 163)	Table 163–11
CAUI-4 C2C (Annex 83D)	Table 83D–6
100GAUI-4 C2C (Annex 135D)	Table 83D–6
100GAUI-2 C2C (Annex 135F)	Table 120D–8
100GAUI-1 C2C (Annex 120F)	Table 120F–8
200GBASE-CR4 (Clause 136)	Table 136–18
200GBASE-CR2 (Clause 162)	Table 162–20
200GBASE-KR4 (Clause 137)	Table 137–6
200GBASE-KR2 (Clause 163)	Table 163–11
200GAUI-8 C2C (Annex 120B)	Table 83D–6 ^a
200GAUI-4 (Annex 120D)	Table 120D–8
200GAUI-2 C2C (Annex 120F)	Table 120F–8
400GBASE-CR4 (Clause 162)	Table 162–20
400GBASE-KR4 (Clause 163)	Table 163–11
400GAUI-16 (Annex 120B)	Table 83D–6 ^a
400GAUI-8 (Annex 120D)	Table 120D–8
400GAUI-4 C2C (Annex 120F)	Table 120F–8

^aWith the exceptions given in 120B.4.

93A.1.2 Transmitter and receiver device package models

93A.1.2.1 Cascade connection of two-port networks

Change the last sentence in 93A.1.2.1 as follows:

For the purpose of this annex, this set of operations is referred to using the shorthand notation $S^{(z)} = \text{cascade}(S^{(x)}, S^{(y)})$. The shorthand notation $S^{(z)} = \text{cascade}(S^{(w)}, S^{(x)}, S^{(y)})$ is equivalent to $S^{(z)} = \text{cascade}(S^{(w)}, \text{cascade}(S^{(x)}, S^{(y)}))$. Following this approach, the notation can be extended to an arbitrary number of components.

93A.1.2.2 Two-port network for a shunt capacitance

Change the second paragraph in 93A.1.2.2 as follows:

The scattering parameters for the device capacitance C_d are denoted as $S^{(d)}=S(C_d)$. the scattering parameters for the device-to-package interface capacitance are denoted as $S^{(b)}=S(C_b)$, and the scattering parameters for the board capacitance C_p are denoted as $S^{(p)}=S(C_p)$. When a value for C_b is not provided by the clause that invokes this method, C_b is set to 0.

Insert 93A.1.2.2a after 93A.1.2.2:

93A.1.2.2a Two-port network for a series inductance

The scattering parameters for a series inductance with value L are defined by Equation (93A–9a).

$$S(L) = \frac{1}{2 + (j\omega L)/R_0} \begin{bmatrix} (j\omega L)/R_0 & 2 \\ 2 & (j\omega L)/R_0 \end{bmatrix} \quad (93A-9a)$$

The scattering parameters for the series inductance L_s are denoted as $S^{(s)}(L_s)$. When a value for L_s is not provided by the clause that invokes this method, L_s is set to 0.

93A.1.2.3 Two-port network for the package transmission line

Change the first paragraph of 93A.1.2.3 as follows:

The scattering parameters for the package transmission line model are a function of the complex propagation coefficient defined by Equation (93A–9), Equation (93A–10), and Equation (93A–11) and the reflection coefficient defined by Equation (93A–12). The values of the parameters that appear in these equations are defined in Table 93A–3 unless alternative values are provided by the clause that invokes this method. Where a value for Z_c is not provided by the clause that invokes this method, it takes the value 78.2 Ω . The units of f are GHz.

Insert the following at the end of 93A.1.2.3:

For clauses that use a second package transmission line segment described by parameters z_{p2} and Z_{c2} , the scattering parameters for the second transmission line are defined by Equation (93A–12), Equation (93A–13), and Equation (93A–14) where values for z_p and Z_c in the equations are provided by z_{p2} and Z_{c2} .

Change the title of Table 93A–3 as follows:

Table 93A–3—Transmission line model parameters and default values

93A.1.2.4 Assembly of transmitter and receiver device package models

Change 93A.1.2.4 as follows:

The scattering parameters for the transmitter device package model $S^{(tp)}$ are the result of the cascade connection of the ~~device capacitance model~~, package transmission line, and board capacitance as shown in Figure 93A–2 and defined by Equation (93A–15) and Equation (93A–15a).

$$\cancel{S^{(tp)} = \text{cascade}(\text{cascade}(S^{(d)}, S^{(l)}), S^{(p)})} \quad S^{(td)} = \text{cascade}(S^{(d)}, S^{(s)}, S^{(b)}) \quad (93A-15)$$

$$\underline{S^{(tp)} = \text{cascade}(S^{(td)}, S^{(l)}, S^{(p)})} \quad (93A-15a)$$

Similarly, the scattering parameters for the receiver device package model $S^{(rp)}$ are the result of the cascade connection of the board capacitance, package transmission line, and ~~device capacitance model~~ as shown in Figure 93A–2 and defined by Equation (93A–16) and Equation (93A–16a).

$$\cancel{S^{(rp)} = \text{cascade}(\text{cascade}(S^{(p)}, S^{(l)}), S^{(d)})} \quad S^{(rd)} = \text{cascade}(S^{(b)}, S^{(s)}, S^{(d)}) \quad (93A-16)$$

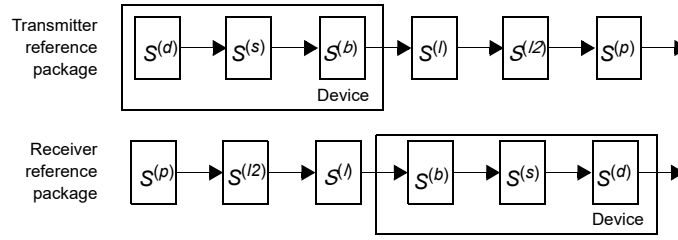
$$\underline{S^{(rp)} = \text{cascade}(S^{(p)}, S^{(l)}, S^{(rd)})} \quad (93A-16a)$$

The transmitter and receiver device package model scattering parameters for clauses that include a second transmission line segment are computed using Equation (93A–16b) and Equation (93A–16c), respectively.

$$\underline{S^{(tp)} = \text{cascade}(S^{(td)}, S^{(l)}, S^{(l2)}, S^{(p)})} \quad (93A-16b)$$

$$\underline{S^{(rp)} = \text{cascade}(S^{(p)}, S^{(l2)}, S^{(l)}, S^{(rd)})} \quad (93A-16c)$$

Insert Figure 93A–2 at the end of 93A.1.2.4 as follows:



NOTE— For some PMDs, $S^{(l/2)}$ is not included in the package model.

$S^{(d)}$ = device capacitance S-parameter
 $S^{(s)}$ = device series inductance S-parameter
 $S^{(b)}$ = bump capacitance S-parameter
 $S^{(l)}$ = package transmission line S-parameter
 $S^{(l/2)}$ = package transmission line 2 S-parameter
 $S^{(p)}$ = package capacitance S-parameter

Figure 93A–2—Reference package models

93A.1.6 Determination of variable equalizer parameters

Change the first paragraph of 93A.1.6 as follows:

COM is a function of the variables $c(-3)$, $c(-2)$, $c(-1)$, $c(1)$, g_{DC} , and g_{DC2} . The following procedure is used to determine the values of these variables that are used to calculate COM.

- Compute the pulse response $h^{(k)}(t)$ of each signal path k for a given $c(-3)$, $c(-2)$, $c(-1)$, $c(1)$, g_{DC} , and g_{DC2} using the procedure defined in 93A.1.5.
- Define t_s to be the time that satisfies Equation (93A–25). If there are multiple values of t_s that satisfy the equation, then the first value prior to the peak of $h^{(0)}(t)$ is selected. The ~~fixed-tap~~ coefficients of the ~~decision feedback equalizer~~ DFE $b(n)$ are computed as shown in Equation (93A–26). If N_b is 0, then the $b(n)$ is considered to be zero for all n . If $bb_{\min}(n)$ and $bb_{\max}(n)$ are not provided by the clause that invokes this method then $bb_{\min}(n)$ is set to $-b_{\max}(n)$ and $bb_{\max}(n)$ is set to $b_{\max}(n)$.
- Compute the location and values of floating-tap coefficients of the DFE to minimize the residual ISI using the method described in 93A.1.6.1. If N_{bg} , N_{bf} , N_f , and $b_{g\max}$ are not specified then no floating taps are used and N_f takes the value of N_b from the referring clause.
- ~~d)~~ e) Define A_s to be $R_{LM}h^{(0)}(t_s)/(L-1)$.
- ~~e)~~ f) Compute σ_{TX}^2 per Equation (93A–30) and Equation (93A–29). This represents the noise output from the transmitter.
- ~~f)~~ g) Compute $h_{ISI}(n)$ per Equation (93A–27). This represents the residual intersymbol interference (ISI) after decision feedback equalization. The corresponding ISI amplitude variance σ_{ISI}^2 is computed per Equation (93A–31) and Equation (93A–29).
- ~~g)~~ h) Compute the slope of the pulse response of the victim path $h_j(n)$ as shown in Equation (93A–28). The variance of the amplitude error due to timing jitter σ_j^2 is computed per Equation (93A–32) and Equation (93A–29).
- ~~h)~~ i) The variance of the amplitude for path k is given by Equation (93A–33) where the phase index m can assume any integer value from 0 to $M-1$. Denote the value of m that maximizes the variance for path k as i . The variance of the amplitude for the combination of all crosstalk paths σ_{XT}^2 is then computed using Equation (93A–34), which is the sum of the maximum variances for the individual paths $k=1$ to $K-1$.

- i) ~~h)~~ Compute the variance of the noise at the output of the receive equalizer σ_N^2 based on the one-sided spectral density η_0 referred to the receiver noise filter input per Equation (93A–35).
- j) ~~i)~~ Compute the figure of merit (FOM) per Equation (93A–36).

Replace Equation (93A–26) in 93A.1.6 with the following equation:

$$b(n) = \begin{cases} bb_{\min}(n) & h^{(0)}(t_s + nT_b)/h^{(0)}(t_s) < bb_{\min}(n) \\ bb_{\max}(n) & h^{(0)}(t_s + nT_b)/h^{(0)}(t_s) > bb_{\max}(n) \\ h^{(0)}(t_s + nT_b)/h^{(0)}(t_s) & \text{otherwise} \end{cases} \quad (93A-26)$$

Replace Equation (93A–27) in 93A.1.6 with the following equation:

$$h_{ISI}(n) = \begin{cases} 0 & n = 0 \\ h^{(0)}(t_s + nT_b) - h^{(0)}(t_s)b(n) & 1 \leq n \leq N_f \\ h^{(0)}(t_s + nT_b) & \text{otherwise} \end{cases} \quad (93A-27)$$

Insert 93A.1.6.1 at the end of 93A.1.6 as follows:

93A.1.6.1 Determination of DFE floating tap locations and magnitudes

The following procedure is used to determine the bank location and tap coefficient magnitudes for each bank of DFE floating taps.

First, the locations of the floating banks are determined using the following steps:

- a) Potential locations for the first tap of a floating bank start at the first post-cursor position after the fixed taps, $N_b + 1$, and end at the specified maximum location minus the number of taps in the bank plus one, $N_f - N_{bf} + 1$. Set the initial values of the provisional coefficients, defined as $b'(n)$, to zero for $N_b + 1 \leq n \leq N_f$.
- b) Define σ_{DFE}^2 to be the sum-of-squares of the tap coefficients for a bank, according to Equation (93A–37a). Calculate σ_{DFE}^2 for each potential bank location, where the coefficient magnitudes are determined using Equation (93A–26). When performing the calculation, if any tap of the floating bank that is being placed overlaps with one or more taps of a previously placed bank, then remove that location from consideration.
- c) Place the floating bank at the location that has the maximum σ_{DFE}^2 . The coefficient magnitudes calculated in b) become the provisional coefficients, $b'(n)$, for the bank, subject to the tail constraints in the steps that follow.
- d) Repeat steps a) through c) for each bank.

After placing the floating banks, the coefficient magnitudes for all floating taps are computed as follows:

- e) For all floating taps that are located between $N_b + 1$ and the start of the ISI tail, $N_{ts} - 1$, set $b(n) = b'(n)$.
- f) For all floating taps that are located between N_{ts} and N_f , compute the tail coefficient root-sum-of-squares, σ_t , which is defined in Equation (93A–37b). Determine the final tap coefficient magnitudes, $b(n)$, by applying the tail weight constraint defined in Equation (93A–37c).

$$\sigma_{DFE}^2 = \sum_{k=1}^{N_{bf}} [b'(N_b + k)]^2 \quad (93A-37a)$$

$$\sigma_t^2 = \sum_{k=N_{ts}}^{N_f} [b'(k)]^2 \quad (93A-37b)$$

$$b(n) = \begin{cases} b'(n) & \sigma_t \leq \sigma_{tmax} \\ b'(n) \frac{\sigma_{tmax}}{\sigma_t} & \sigma_t > \sigma_{tmax} \end{cases} \quad (93A-37c)$$

93A.5 Effective Return Loss (ERL)

Change Table 93A-4 as follows (some unchanged rows not shown):.

Table 93A-4—ERL parameters

Parameter	Reference	Symbol	Units
...			
Target detector error ratio	93A.1.7	DER_0	—
Time-gated propagation delay	93A.5.2	T_{fx}	ns
Tukey window flag	93A.5.1	tw	—

93A.5.1 Pulse time-domain reflection signal

Change the second paragraph of 93A.5.1 as follows:

The filtered return loss, $H_{ii}(f)$, is defined by Equation (93A-58). When the parameter tw is equal to 1, $H_{tw}(f)$ is defined by Equation (93A-58a). When the value of tw is 0 or is not provided by the clause that invokes this method, $H_{tw}(f) = 1$.

Replace Equation (93A-58) with the following equation:

$$H_{ii}(f) = H_i(f)s_{ii}(f)H_r(f)H_{tw}(f) \quad (93A-58)$$

Insert new Equation (93A-58a) and Equation (93A-58b) after Equation (93A-58):

$$H_{tw}(f) = \begin{cases} 1 & f \leq f_r \\ 0.5 \left(1 - \cos \left(\frac{2\pi(f-f_b)}{f_{per}} \right) \right) & f_r < f \leq f_b \\ 0 & f_b < f \end{cases} \quad (93A-58a)$$

$$f_{per} = 2(f_b - f_r) \quad (93A-58b)$$

Change the variable list after Equation (93A-58), now after Equation (93A-58b), in 93A.5.1 as follows:

where

f	is the frequency in GHz
$H_r(f)$	is defined by Equation (93A-20)
$H_t(f)$	is defined by Equation (93A-46)
f_{per}	<u>is defined by Equation (93A-58b)</u>
i	is the port index of the scattering parameters, 1 or 2
f_r	<u>is the receiver 3 dB bandwidth in GHz</u>
$H_{tw}(f)$	<u>is the Tukey window function</u>

93A.5.2 Effective reflection waveform

Change the definition of T_{fx} in the variable list after Equation (93A-62) as follows:

T_{fx}	is twice the propagation delay in ns associated with the test fixture, obtained by measurement or inspection, <u>unless its value is specified by the clause that invokes this method</u>
----------	---

Annex 120A

(informative)

200 Gb/s and 400 Gb/s PMA sublayer partitioning examples

Insert new subclause 120A.5 at the end of Annex 120A as follows:

120A.5 Partitioning examples supporting 200GBASE-CR2/KR2 and 400GBASE-CR4/KR4

Figure 120A–8 depicts an example of 200GBASE-CR2/KR2 and 400GBASE-CR4/KR4 PMA layering with a single 200GAUI-*n* or 400GAUI-*n* interface.

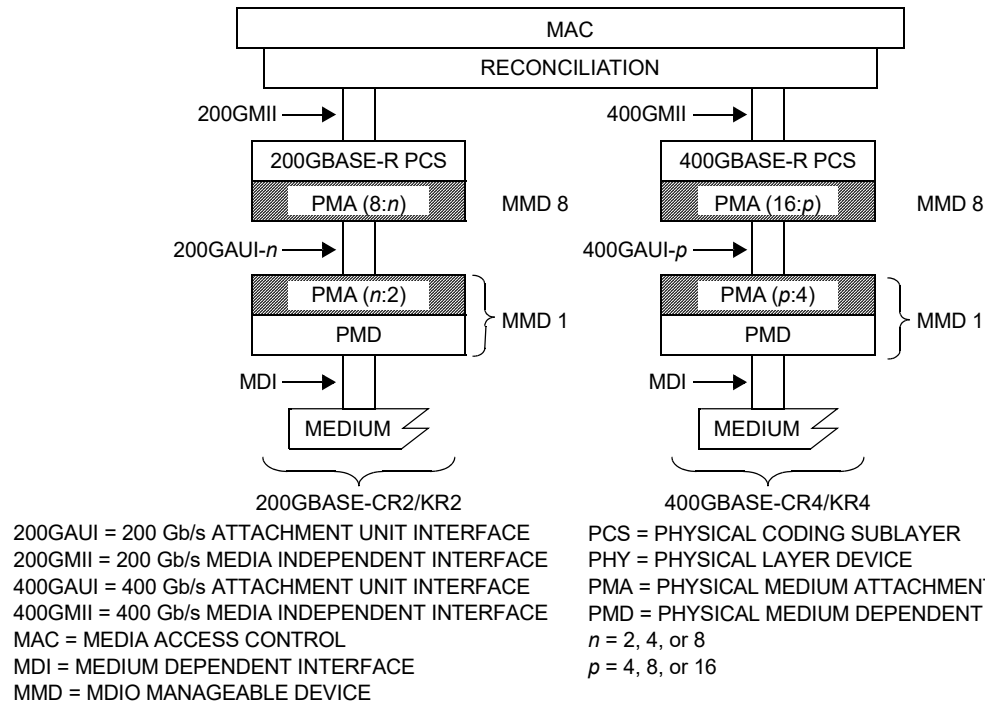


Figure 120A–8—Example 200GBASE-CR2/KR2 and 400GBASE-CR4/KR4 PMA layering with single 200GAUI-2/4/8 or 400GAUI-4/8/16 C2C interface

Insert new annexes Annex 120F and Annex 120G corresponding to Clause 120 as follows:

Annex 120F

(normative)

Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)

120F.1 Overview

This annex defines the functional and electrical characteristics for the optional chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C). The 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C interfaces provide electrical characteristics and associated compliance points, which can optionally be used when designing systems with electrical interconnect of approximately 25 cm in length.

100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C are physical instantiations of the PMA service interface between the FEC (for 100GBASE-P PHYs) or PCS (for 200GBASE-R and 400GBASE-R PHYs) and the PMD, as described in 135.1.4 and 120.1.4. Figure 120F–1 shows example relationships of the 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C interfaces to the ISO/IEC Open System Interconnection (OSI) reference model.

The sublayers including PCS and FEC (for 100GBASE-P PHYs) for each PHY that can optionally include a 100GAUI-1 C2C, 200GAUI-2 C2C, or 400GAUI-4 C2C are summarized in the tables in 80.1.5 and 116.1.4 and are specified in the corresponding PMD clause. The positioning of the 100GAUI-1 C2C, 200GAUI-2 C2C, or 400GAUI-4 C2C relative to other sublayers is shown in 135.1 and 120.1 with further examples in Annex 135A and Annex 120A.

The 100GAUI-1, 200GAUI-2, or 400GAUI-4 C2C bidirectional link is described in terms of a C2C transmitter, a C2C channel, and a C2C receiver. Figure 120F–2 depicts a typical C2C application. The C2C components at both ends of the link have connected ground references.

The C2C interface is composed of independent data paths in each direction. Each 100GAUI-1, 200GAUI-2, or 400GAUI-4 data path contains one, two, or four, respectively, differential lanes, which are AC-coupled.

The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C use PAM4 signaling. The nominal signaling rate for each lane is 53.125 GBd.

The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C channel is specified in 120F.4.

The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C transmitter on each end of the link is adjusted to an appropriate setting, which may be based on channel knowledge. If implemented, the transmitter equalization feedback mechanism described in 120F.3.2.6 may be used to identify an appropriate setting of the link partner transmitter. The adaptive or adjustable receiver performs the remainder of the equalization.

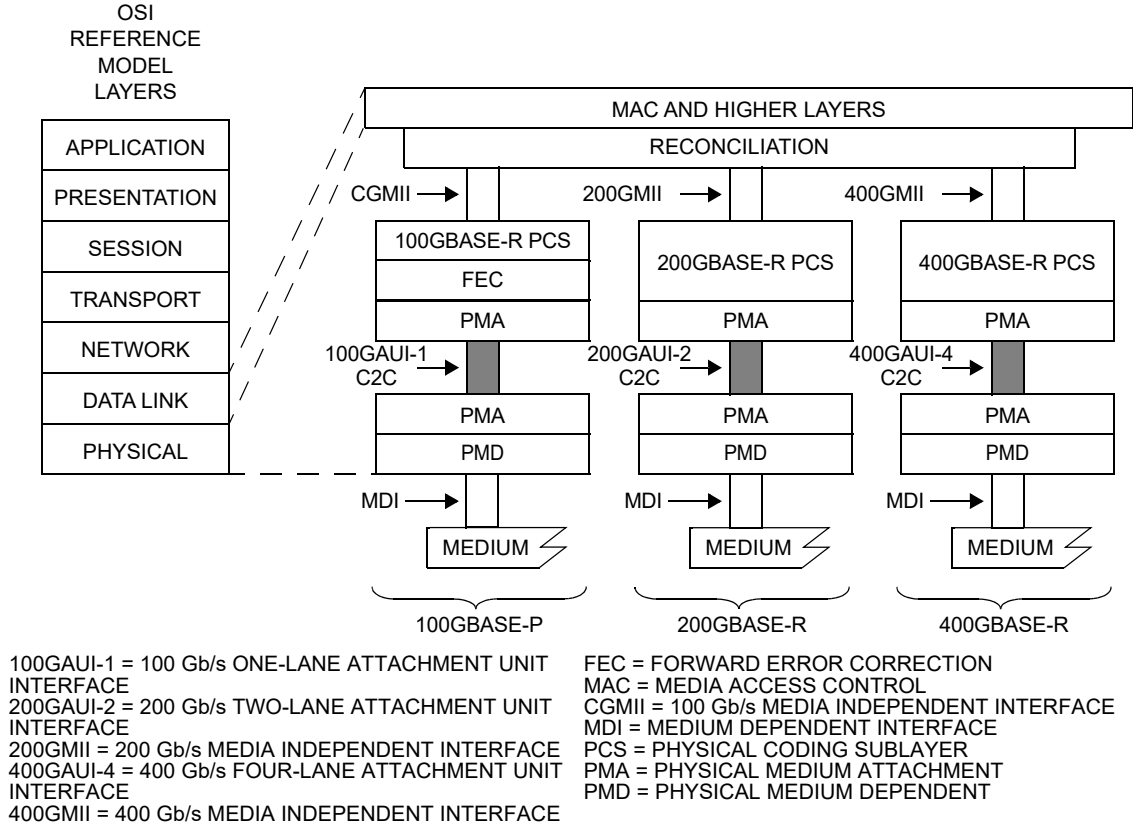


Figure 120F-1—Example 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

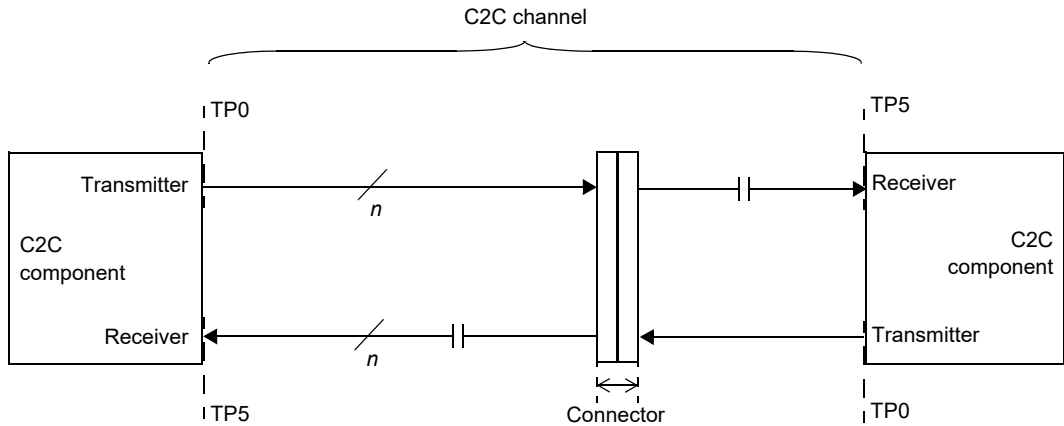


Figure 120F-2—Typical 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C application

The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C transmitter supports $1/(1+D)$ mod 4 precoding, as specified in 135.5.7.2 and 120.5.7.2, that may be enabled or disabled as required. The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C receiver may support $1/(1+D)$ mod 4 precoding, as specified in 135.5.7.2 and 120.5.7.2. Precoding may be enabled and disabled using the precoder request mechanism specified in 135F.3.2.1.

120F.2 Compliance points

The electrical characteristics for the 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C interfaces are defined at test points as described in 163.9.2.1 and 163.9.3.2.

120F.2.1 Reference impedance

The reference impedance for differential specifications is $100\ \Omega$. The reference impedance for common-mode specifications is $25\ \Omega$.

120F.3 Electrical characteristics

120F.3.1 Transmitter electrical characteristics

A 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C transmitter shall meet the specifications given in Table 120F–1.

Table 120F–1—Transmitter electrical characteristics at TP0v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)		$53.125 \pm 50\ \text{ppm}^a$	GBd
Differential peak-to-peak output voltage ^b (max) Transmitter disabled Transmitter enabled	93.8.1.3	35 1200	mV mV
Common-mode voltage ^b (max)	93.8.1.3	1	V
Common-mode voltage ^b (min)	93.8.1.3	0.2	V
Low-frequency peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ (max)	120F.3.1.1	32	mV
Signal to AC common-mode noise ratio, $SCMR$ (min)	120F.3.1.2	15	dB
Difference effective return loss, $dERL$ (min)	120F.3.1.4	–3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	162.9.4.9	2	dB
Difference steady-state voltage, dv_f (min)	163.9.2.4	0	V
Difference linear fit pulse peak ratio, dR_{peak} (min)	163A.3.2.1	0	—
Level separation mismatch ratio, R_{LM} (min)	162.9.4.2	0.95	—

Table 120F–1—Transmitter electrical characteristics at TP0v (*continued*)

Parameter	Reference	Value	Units
Output waveform ^c			
absolute value of step size for all taps (min)	162.9.4.1.4	0.005	—
absolute value of step size for all taps (max)	162.9.4.1.4	0.025	—
value at min state for c(−3) (max)	162.9.4.1.5	−0.05	—
value at max state for c(−3) (min)	162.9.4.1.5	0	—
value at min state for c(−2) (max)	162.9.4.1.5	0	—
value at max state for c(−2) (min)	162.9.4.1.5	0.1	—
value at min state for c(−1) (max)	162.9.4.1.5	−0.3	—
value at max state for c(−1) (min)	162.9.4.1.5	0	—
value at min state for c(0) (max)	162.9.4.1.5	0.5	—
value at min state for c(1) (max)	162.9.4.1.5	−0.1	—
value at max state for c(1) (min)	162.9.4.1.5	0	—
Signal-to-noise-and-distortion ratio, SNDR (min)	162.9.4.6	32.5	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	120F.3.1.3	28	dB
Output jitter			
J_{RMS} (max)	120F.3.1.6	0.023	UI
J_{4u} (max)	120F.3.1.6	0.128	UI
$J_{4u_{03}}$ (max)	120F.3.1.6	0.118	UI
Even-odd jitter (max)	120F.3.1.6	0.025	UI

^a For a PMA in the same package as the PCS sublayer. In other cases, the signaling rate is derived from the signaling rate presented to the input lanes (see Figure 135–3 and Figure 120–3) by the adjacent PMD, PMA, or FEC sublayers.

^b Measurement uses the method described in 93.8.1.3 with the exception that the PRBS13Q test pattern is used.

^c The state of the transmit equalizer is controlled by management interface.

A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for all transmitter signal measurements, unless otherwise specified.

120F.3.1.1 Peak-to-peak AC common-mode voltage

The low-frequency and full-band peak-to-peak AC common-mode voltage, VC_{MF} and VC_{MB} , respectively, are defined by the method specified in 162.9.4.4 with the exception that the peak-to-peak AC common-mode voltage is defined as the AC common-mode voltage range measured at TP0v that includes all but 10^{-5} of the measured distribution, from 0.000005 to 0.999995 of the cumulative distribution.

The low-frequency peak-to-peak AC common-mode voltages shall meet the specification for VC_{MF} (max) in Table 120F–1.

120F.3.1.2 Signal to AC common-mode noise ratio

Signal to AC common-mode noise ratio, $SCMR$, is defined by the method specified in 163.9.2.6 with the exception that the full-band peak-to-peak AC common-mode voltage is defined in 120F.3.1.1.

The signal to AC common-mode noise ratio shall meet the specification for $SCMR$ (min) in Table 120F–1.

120F.3.1.3 Signal-to-residual-intersymbol-interference ratio

Signal-to-residual-intersymbol-interference ratio SNR_{ISI} is defined by the method specified in 162.9.4.3 with the exception that the continuous time filter settings are provided in Table 120F–8.

Signal-to-residual-intersymbol-interference ratio shall meet the specification for SNR_{ISI} (min) in Table 120F–1.

120F.3.1.4 Transmitter difference effective return loss (ERL)

The transmitter difference ERL at TP0v is computed using the procedure in 163A.3.2.2 with the values in Table 120F–2 and Table 120F–8, and with the value of T_{fx} equal to twice the delay from TP0 to TP0v.

Table 120F–2—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	200	UI
Equalizer length associated with reflection signal	N_{bx}	6	UI
Tukey window flag	tw	1	—

The difference ERL at TP0v shall be greater than or equal to $dERL$ (min) specified in Table 120F–1.

120F.3.1.5 Transmitter equalization settings

Each transmitter includes programmable equalization to compensate for the frequency-dependent loss of the channel and to facilitate data recovery at the receiver. The functional model for the transmit equalizer is the five-tap transversal filter shown in Figure 120F–3. Transmitter output waveform specifications are provided in Table 120F–1. The state of the transmitter equalizer may be configured via the transmitter control interface described in 120F.3.1.7.

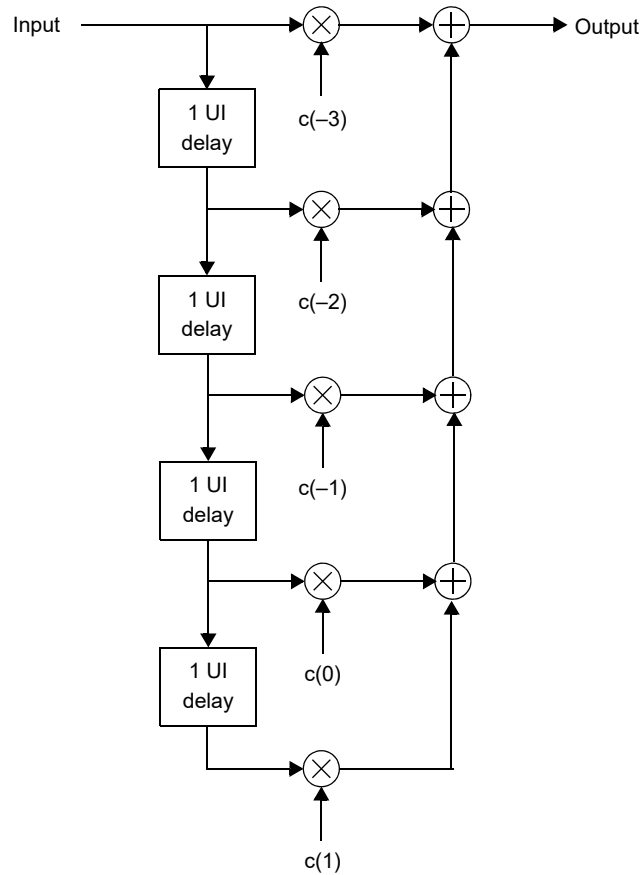


Figure 120F-3—Transmitter equalizer functional model

120F.3.1.6 Output jitter

Output jitter is characterized by four parameters: J_{4u} , $J_{4u_{03}}$, J_{RMS} , and even-odd jitter. These parameters are calculated from measurements with a single transmit equalizer setting to compensate for the loss of the transmitter package and test fixture. The equalizer setting is chosen to minimize any or all of the jitter parameters.

J_{4u} and J_{RMS} are calculated using the jitter measurement method specified in 120D.3.1.8.1. $J_{4u_{03}}$ is calculated the same way as J_{4u} except that the calculation uses only transitions R03 and F30 in Table 120D-4.

Even-odd jitter is calculated using the jitter measurement method specified in 162.9.4.7.

NOTE— J_{4u} is sensitive to measurement noise being converted to timing errors. Hence, accounting for measurement noise effects is recommended.

120F.3.1.7 Transmitter control

The transmitter output equalization is characterized using the linear fit method described in 162.9.4.1.1.

Transmitter equalization shall be configurable by management using a set of control and status variables listed in Table 120F–3, or an equivalent interface. The variables that control transmitter equalization are specific for each lane.

Table 120F–3—Transmitter control variables

Variable name	Description	Management access ^a
local_ic_req	Requested initial condition in the local transmitter.	RW
local_coef_sel	Coefficient selection in the local transmitter.	RW
local_coef_req	Update to the selected coefficient in the local transmitter.	RW
local_ic_sts	Status of initial condition request.	RO
local_k	Coefficient select echo.	RO
local_coef_sts	Coefficient update acknowledge.	RO

^a RW = Read/Write, RO = Read only.

Definitions of the variables in Table 120F–3 are the same as the corresponding ones without the “local_” prefix in 136.8.11.7.1, with the exceptions listed in 162.8.11, and the additional exception that all variables are accessed and controlled by management instead of being mapped to training frame fields.

If a Clause 45 MDIO is implemented, the variable mapping for the transmitter control is identical to the “LD” variable mapping in Table 162–7.

120F.3.2 Receiver characteristics

A receiver shall meet the specifications given in Table 120F–4, with precoding enabled on the test transmitter if required by the receiver.

Table 120F–4—Receiver characteristics at TP5v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)	120F.3.2.1	53.125 ± 100 ppm	GBd
Difference effective return loss, <i>dERL</i> (min)	120F.3.2.2	–3	dB
Differential-mode to common-mode return loss, <i>RLcd</i>	120F.3.2.3	Equation (120F–1)	dB
Interference tolerance	120F.3.2.4	Table 120F–5	—
Jitter tolerance	120F.3.2.5	Table 162–17	—

120F.3.2.1 Receiver signaling rate

The receiver shall comply with the requirements of 120F.3.2.4 and 120F.3.2.5 for any signaling rate in the range $53.125 \text{ GBd} \pm 100 \text{ ppm}$.

120F.3.2.2 Receiver difference ERL

The receiver difference ERL at TP5v is computed using the procedure in 163A.4.1.2 with the values in Table 120F–2 and Table 120F–8, and with the value of T_{fx} equal to twice the delay from TP5 to TP5v.

Receiver difference ERL at TP5v shall be greater than or equal to $dERL$ (min) specified in Table 120F–4.

120F.3.2.3 Receiver differential-mode to common-mode return loss

The differential-mode to common-mode return loss of the receiver shall meet Equation (120F–1). The differential-mode to common-mode return loss limit is illustrated by Figure 120F–4.

$$RL_{cd}(f) \geq \begin{cases} 25 - 0.72f & 0.05 \leq f \leq 13.9 \\ 15 & 13.9 < f \leq 38 \end{cases} \quad (120F-1)$$

where

RL_{cd} is the differential-mode to common-mode return loss in dB
 f is the frequency in GHz

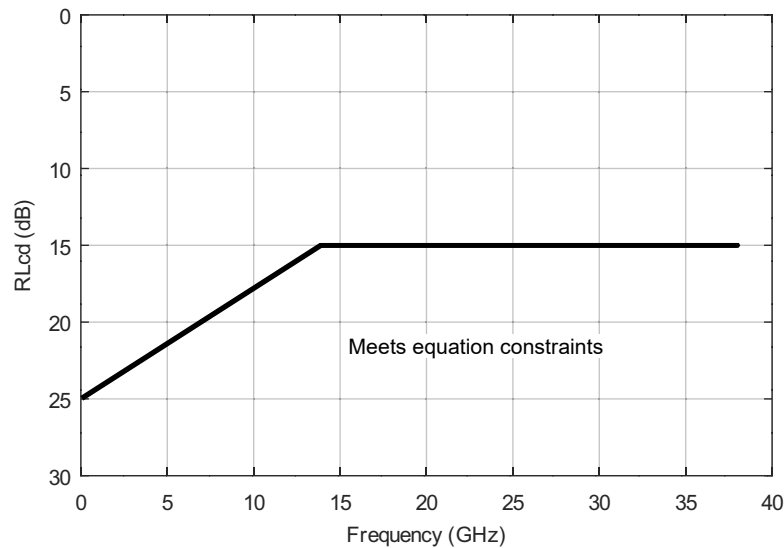


Figure 120F–4—Receiver differential-mode to common-mode return loss limit

120F.3.2.4 Receiver interference tolerance

Receiver interference tolerance is defined by the procedure in [Annex 93C](#) with the exception that transmitter equalization is configured by management (see [120D.3.2.3](#)) to the settings that provide the lowest FEC symbol error ratio. The receiver on each lane shall meet the FEC symbol error ratio requirement with

channels matching the Channel Operating Margin (COM) and loss parameters for Test 1 and Test 2 in Table 120F–5. The following additional considerations apply to the interference tolerance test.

- a) The test transmitter is constrained such that for any transmitter equalizer setting the differential peak-to-peak voltage (see 93.8.1.3) is less than or equal to 800 mV.
- b) The ERL of the test channel measured at TP5 replica towards TPt meets the requirements in 120F.4.3.
- c) The lower frequency bound for the noise spectral density constraints, f_{NSDI} , is 1 GHz.
- d) For the calculation of COM, the transmitter model is determined according to step e in 163.9.3.5.
- e) For the calculation of test channel COM, the following parameters are based on values measured from the test transmitter.
 - The parameter SNR_{TX} is set to the measured value of SNDR with $N_p=11$.
 - The parameter R_{LM} is set to the measured value of R_{LM} .
 - The parameters A_{DD} and σ_{RJ} are calculated from the measured values of J_{4u} and J_{RMS} using Equation (120D–10), and Equation (120D–11), respectively, with the exceptions that $Q4$ is replaced with $Q4d$ and $Q4d$ is equal to 3.719.
- f) Other COM parameters are set according to the values in Table 120F–8.
- g) COM is calculated using both Test 1 and Test 2 device package model transmission line lengths listed in Table 120F–8 on the receiver side. The value of COM is taken as the lower of the two calculated values.
- h) The test pattern is the scrambled idle test pattern defined in 119.2.4.9.
- i) As an alternative to using the scrambled idle test pattern and measuring FEC symbol error ratio it is permissible to use the PRBS31Q pattern as described in 120.5.11.2.2 and bit error ratio testing. In this case the required bit error ratio is equal to the required FEC symbol error ratio divided by 10. Note that this requirement can be somewhat more stringent than using the scrambled idle test pattern and measuring FEC symbol error ratio, and therefore failing this test requirement with the PRBS31Q pattern does not necessarily imply a failure of the interference tolerance test.
- j) A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for measurement of the signal applied by the pattern generator and for measurements of the broadband noise.

Table 120F–5—Receiver interference tolerance parameters

Parameter	Test 1 (low loss)			Test 2 (high loss)			Units
	Min	Max	Target	Min	Max	Target	
FEC symbol error ratio ^a	—	10^{-4}	—	—	10^{-4}	—	—
Insertion loss at 26.5625 GHz ^b	9.5	10.5	—	19.5	20.5	—	dB
RSS_DFE4 ^c	0.05	—	—	0.05	—	—	—
COM including effects of broadband noise ^d	—	—	3	—	—	3	dB

^a The FEC symbol error ratio is measured in step 10) of the receiver interference tolerance method defined in 93C.2

^b Measured between TPt and TP5 (see Figure 93C–4).

^c RSS_DFE4 is described in 93A.2.

^d The COM value is the target for the receiver noise level calibration defined in 93C.2 step 7). The channel noise voltage applied in 93C.2 step 8) should be as close as practical to the value needed to produce the target COM. If higher amplitude values are used, this would demonstrate margin to the specification but this is not required for compliance.

120F.3.2.5 Receiver jitter tolerance

Receiver jitter tolerance is verified for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 162–17. The test setup shown in Figure 93–12, or its equivalent, is used. The test channel meets the insertion loss requirement for Test 2 in Table 120F–5. The synthesizer frequency is set to the specified jitter frequency and the synthesizer output amplitude is adjusted until the specified peak-to-peak jitter amplitude for that frequency is measured at TP0v. The test procedure is the same as the one described in 120F.3.2.4, with the following exceptions:

- No broadband noise is added.
- The test channel COM, calculated per items d) through g) in 120F.3.2.4, is at least 3 dB.
- For the COM parameter calibration described in 120F.3.2.4 item e), the test channel transmitter J_{RMS} and J_{4u} values are measured with the jitter frequency and amplitude set according to Case F from Table 162–17.

The receiver under test shall meet the FEC symbol error ratio in Table 120F–5 for each case in Table 162–17.

120F.3.2.6 Transmitter equalization feedback (optional)

Transmitter equalization feedback is an optional capability for a receiver. If implemented, it shall operate as described in this subclause.

A receiver may generate a request to change the transmit equalization coefficients of the remote transmitter using the variables in Table 120F–6, or an equivalent interface.

Table 120F–6—Transmitter equalization feedback variables

Variable name	Description	Management access ^a
remote_ic_req	Initial condition request for the remote transmitter.	RO
remote_coef_sel	Coefficient selection for the remote transmitter.	RO
remote_coef_req	Requested update to the selected coefficient for the remote transmitter.	RO
remote_ic_sts	Status of initial condition request from the remote transmitter.	RW
remote_k	Coefficient select echo from the remote transmitter.	RW
remote_coef_sts	Coefficient update acknowledge from the remote transmitter.	RW

^a RW = Read/Write, RO = Read only.

Definitions of the variables in Table 120F–6 are the same as the corresponding ones without the “remote_” prefix in 136.8.11.7.1, with the exceptions listed in 162.8.11, and the additional exception that all variables are accessed and controlled by management instead of being mapped to training frame fields.

The requested setting is generated from information internal to the receiver in an implementation specific manner.

If a Clause 45 MDIO is implemented, the variable mapping for the transmitter equalization feedback is identical to the “LP” variable mapping in Table 162–7.

120F.4 Channel characteristics

Table 120F–7 provides a summary of the channel characteristics for 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C interfaces, and references to the subclauses addressing each parameter.

Table 120F–7—Channel characteristics summary

Description	Reference	Value	Unit
Minimum COM	120F.4.1	3	dB
Maximum insertion loss at 26.56 GHz (recommended)	120F.4.2	20	dB
Minimum ERL	120F.4.3	9.7	dB
Differential-mode to common-mode return loss, RL_{cd}	120F.4.4	Equation (163–6)	dB
Maximum AC-coupling 3 dB corner frequency	120F.4.5	50	kHz

120F.4.1 Channel operating margin

The Channel Operating Margin (COM) is computed using the procedure in 93A.1, where T_r is 7.5 ps for $H_f(f)$ in Equation (93A–19), and the parameters in Table 120F–8. COM is calculated twice, Test 1 and Test 2, with different values of device package model transmission line length z_p . COM shall be greater than or equal to 3 dB for both Test 1 and Test 2. This minimum value allocates margin for practical limitations on the receiver implementation as well as the allowed transmitter equalization coefficients.

Table 120F–8—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	53.125	GBd
Maximum start frequency	$f_{\min}$	0.05	GHz
Maximum frequency step	Δf	0.01	GHz
Device package model			
Single-ended device capacitance	C_d	$1.2\text{e-}4$	nF
Single-ended device series inductance	L_s	0.12	nH
Single-ended bump capacitance	C_b	3×10^{-5}	nF
Transmission line length, Test 1 Tx	z_p	13	mm
Transmission line length, Test 1 Rx	z_p	11	mm
Transmission line length, Test 2 Tx	z_p	31	mm
Transmission line length, Test 2 Rx	z_p	29	mm
Transmission line 2 length	z_{p2}	1.8	mm
Transmission line parameter, a_1	a_1	9.909×10^{-4}	ns ^{1/2} /mm
Transmission line parameter, a_2	a_2	2.772×10^{-4}	ns/mm
Single-ended package capacitance at package-to-board interface	C_p	8.7×10^{-5}	nF
Package transmission line nominal characteristic impedance	Z_c	87.5	Ω
Package transmission line 2 nominal characteristic impedance	Z_{c2}	92.5	Ω
Single-ended reference resistance	R_0	50	Ω
Single-ended termination resistance	R_d	50	Ω
Receiver 3 dB bandwidth	f_r	$0.75 \times f_b$	GHz

Table 120F–8—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Transmitter equalizer, minimum cursor coefficient	$c(0)$	0.54	—
Transmitter equalizer, 3 rd pre-cursor coefficient Minimum value Maximum value Step size	$c(-3)$	−0.04 0 0.02	—
Transmitter equalizer, 2 nd pre-cursor coefficient Minimum value Maximum value Step size	$c(-2)$	0 0.1 0.02	—
Transmitter equalizer, 1 st pre-cursor coefficient Minimum value Maximum value Step size	$c(-1)$	−0.28 0 0.02	—
Transmitter equalizer, post-cursor coefficient Minimum value Maximum value Step size	$c(1)$	−0.1 0 0.02	—
Continuous time filter, DC gain Minimum value Maximum value Step size	g_{DC}	−20 0 1	dB dB dB
Continuous time filter, DC gain 2 Minimum value Maximum value Step size	g_{DC2}	−4 0 1	dB dB dB
Continuous time filter, zero frequency for $g_{DC} = 0$	f_z	$f_b / 2.5$	GHz
Continuous time filter, pole frequencies	f_{p1} f_{p2}	$f_b / 2.5$ f_b	GHz GHz
Continuous time filter, low-frequency pole/zero	f_{LF}	$f_b / 80$	GHz
Transmitter differential peak output voltage Victim Far-end aggressor Near-end aggressor	A_v A_{fe} A_{ne}	0.413 0.413 0.608	V V V
Number of signal levels	L	4	—
Level separation mismatch ratio	R_{LM}	0.95	—
Transmitter signal-to-noise ratio	SNR_{TX}	33	dB
Number of samples per unit interval	M	32	—
Decision feedback equalizer (DFE) length	N_b	6	UI
Normalized DFE coefficient maximum limit $n = 1$ $n = 2$ $n = 3$ to N_b	$bb_{\max}(n)$	0.65 0.15 0.1	—
Normalized DFE coefficient minimum limit $n = 1$ $n = 2$ $n = 3$ to N_b	$bb_{\min}(n)$	0.3 0.05 −0.04	—

Table 120F-8—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI
One-sided noise spectral density	η_0	2×10^{-8}	V ² /GHz
Target detector error ratio	DER_0	10^{-5}	—

120F.4.2 Channel insertion loss (recommended)

The channel insertion loss should meet Equation (120F-2). Actual channel insertion loss could be higher or lower than that given by Equation (120F-2) due to the channel insertion loss deviation (ILD), return loss, and crosstalk. Note that for this equation the channel insertion loss at the Nyquist frequency is less than or equal to 20 dB.

$$ILdd(f) \leq 1.083 + 1.444\sqrt{f} + 0.432f \quad (120F-2)$$

for $0.01 \leq f \leq 53.125$

where

f is the frequency in GHz
 $ILdd$ is the channel insertion loss in dB

The insertion loss limit is illustrated by Figure 120F-5.

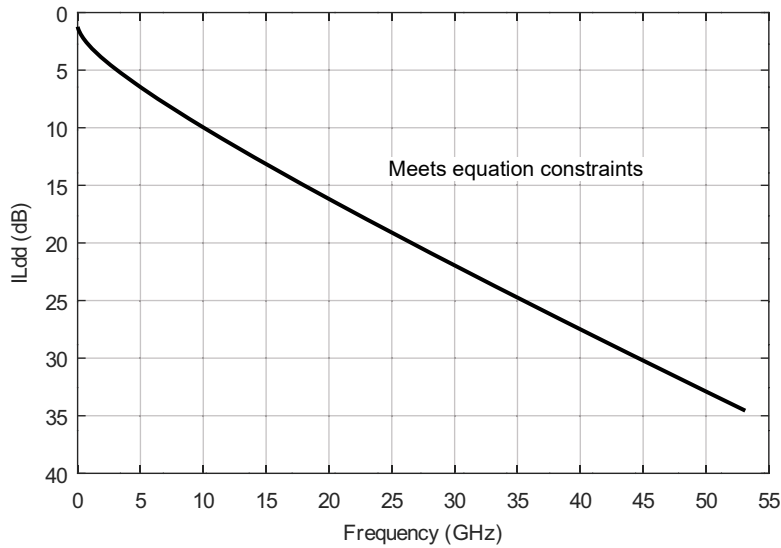


Figure 120F-5—Channel insertion loss limit

120F.4.3 Channel ERL

The channel ERL at TP0 and at TP5 is computed using the procedure in 93A.5 with the values in Table 120F–9. Parameters that do not appear in Table 120F–9 take values from Table 120F–8. Channel ERL at TP0 and at TP5 shall be greater than or equal to 9.7 dB.

Table 120F–9—Channel ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	2000	UI
Equalizer length associated with reflection signal	N_{bx}	6	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

120F.4.4 Channel differential-mode to common-mode return loss

The TP0 to TP5 channel differential-mode to common-mode return loss shall meet Equation (163–6) illustrated in Figure 163–7.

120F.4.5 Channel AC-Coupling

Channel AC-coupling is specified in 163.10.7.

120F.5 Protocol implementation conformance statement (PICS) proforma for Annex 120F, Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)²⁰

120F.5.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 120F, Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

120F.5.2 Identification

120F.5.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

120F.5.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Annex 120F, Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
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²⁰Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

120F.5.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
NOL	Number of differential AC-coupled lanes	120F.1	One independent data path in each direction for 100GAUI-1 C2C, two independent data paths in each direction for 200GAUI-2 C2C, and four independent data paths in each direction for 400GAUI-4 C2C	M	Yes []
*CHAN	Channel	120F.4	Items marked with CHAN include channel specifications not applicable to a PHY manufacturer	O	Yes [] No []
*PCSP KG	PMA in same package as PCS.			O	Yes [] No []

120F.5.4 PICS proforma tables for Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)

120F.5.4.1 Transmitter

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Signaling rate, each lane, PMA is in same package as PCS	120F.3.1	Per Table 120F–1	PCSP KG:M	Yes [] N/A []
TC2	Signaling rate, each lane, PMA is not in same package as PCS	120F.3.1	Derived from PMA input lanes	!PCSP KG:M	Yes [] N/A []
TC3	Peak-to-peak differential output voltage, transmitter enabled	120F.3.1	Per Table 120F–1	M	Yes []
TC4	Peak-to-peak differential output voltage, transmitter disabled	120F.3.1	Per Table 120F–1	M	Yes []
TC5	DC common-mode voltage	120F.3.1	Per Table 120F–1	M	Yes []
TC6	Low-frequency peak-to-peak AC common-mode voltage	120F.3.1	Per Table 120F–1	M	Yes []
TC7	Signal to AC common-mode noise ratio	120F.3.1	Per Table 120F–1	M	Yes []
TC8	Difference ERL	120F.3.1	Per Table 120F–1	M	Yes []
TC9	Common-mode to common-mode return loss	120F.3.1	Per Table 120F–1	M	Yes []
TC10	Output waveform coefficient limits	120F.3.1	Per Table 120F–1	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TC11	Output waveform coefficient step size	120F.3.1	Per Table 120F–1	M	Yes []
TC12	Output jitter	120F.3.1	Per Table 120F–1	M	Yes []
TC13	Signal-to-noise-and-distortion ratio	120F.3.1	Per Table 120F–1	M	Yes []
TC14	Signal-to-residual-intersymbol-interference ratio	120F.3.1	Per Table 120F–1	M	Yes []
TC15	Transmitter precoder request	120F.1		O	Yes [] No []
TC16	Transmit equalizer controlled by management interface	120F.3.1.7		M	Yes []
TC17	Difference steady-state voltage	120F.3.1	Per Table 120F–1	M	Yes []
TC18	Difference linear fit pulse peak ratio	120F.3.1	Per Table 120F–1	M	Yes []
TC19	Level separation mismatch ratio	120F.3.1	Per Table 120F–1	M	Yes []

120F.5.4.2 Receiver

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	Difference ERL	120F.3.2	Per Table 120F–4	M	Yes []
RC2	Differential-mode to common-mode return loss	120F.3.2	Per Table 120F–4	M	Yes []
RC3	Interference tolerance	120F.3.2.4		M	Yes []
RC4	Jitter tolerance	120F.3.2.5		M	Yes []
RC5	Transmitter equalization feedback	120F.3.2.6		O	Yes [] No []

Channel

Item	Feature	Subclause	Value/Comment	Status	Support
CC1	Channel operating margin (COM)	120F.4.1	≥ 3 dB	CHAN :M	Yes [] N/A []
CC2	Channel insertion loss	120F.4.2	Meets Equation (120F–2) constraints	CHAN :O	Yes [] No [] N/A []
CC2	Channel ERL	120F.4.3	≥ 9.7 dB	CHAN :M	Yes [] N/A []
CC4	Channel AC-coupling	120F.4.5	Per 163.10.7	CHAN :M	Yes [] N/A []

Annex 120G

(normative)

Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)

120G.1 Overview

This annex defines the functional and electrical characteristics for the optional chip-to-module (C2M) 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M). Figure 120G–1 shows the relationship of the 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M interfaces to the ISO/IEC Open System Interconnection (OSI) reference model. The C2M interface provides electrical characteristics and associated compliance points, which can optionally be used when designing systems with pluggable module interfaces.

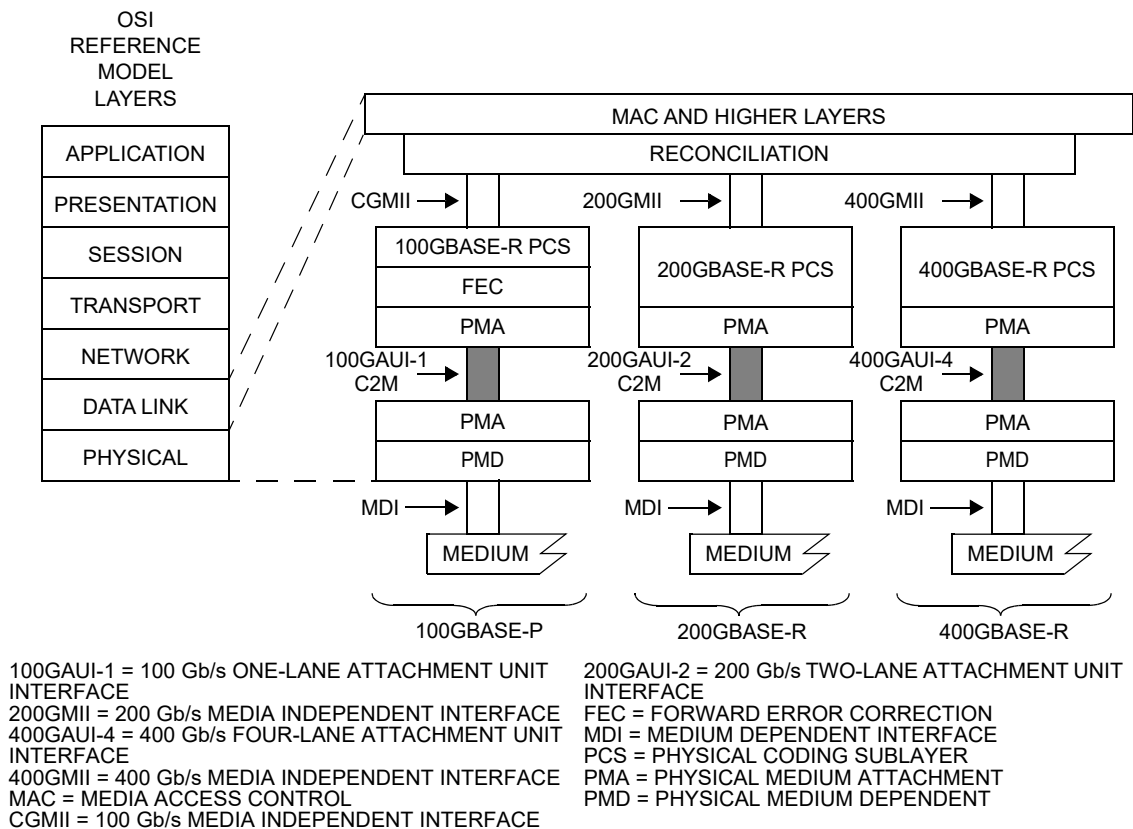
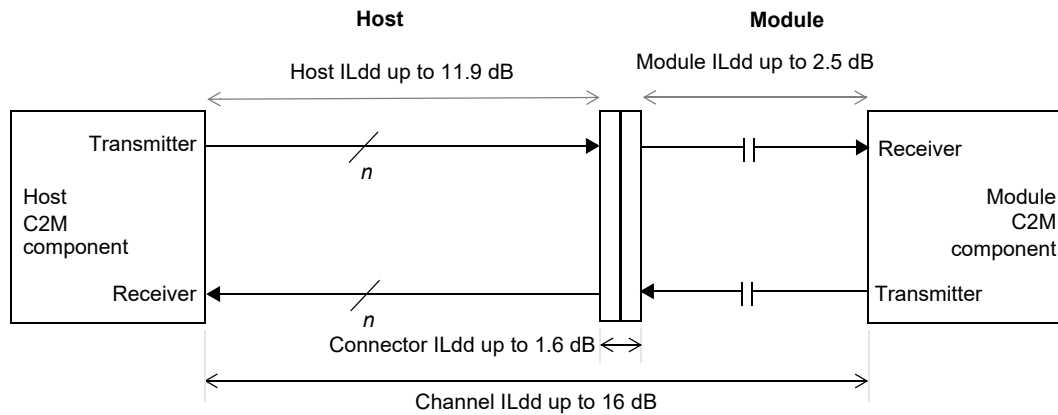


Figure 120G–1—Example 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

The sublayers including PCS and FEC (for 100GBASE-P PHYs) for each PHY that can optionally include a 100GAUI-1 C2M, 200GAUI-2 C2M, or 400GAUI-4 C2M are summarized in the tables in 80.1.5 and 116.1.4 and are specified in the corresponding PMD clause. The positioning of the 100GAUI-1 C2M, 200GAUI-2 C2M, or 400GAUI-4 C2M relative to other sublayers is shown in 135.1 and 120.1 with further examples in Annex 135A and Annex 120A.

The C2M link is described in terms of a host C2M component, a C2M channel with associated insertion loss (ILdd), and a module C2M component. The host C2M component and module C2M component have connected ground references. Figure 120G–2 depicts a typical C2M application and summarizes the ILdd budget associated with the C2M application. The ILdd budget reflects the highest ILdd intended to be supported and is characterized by Equation (120G–5) and illustrated in Figure 120G–12. The C2M interface comprises independent data paths in each direction. Each 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M data path contains one, two, or four differential lanes, respectively, using PAM4 signaling, where the highest differential level corresponds to the symbol three and the lowest level corresponds to the symbol zero. Each lane is AC-coupled within the module.



NOTE—The number of lanes n is equal to 1 for 100GAUI-1, 2 for 200GAUI-2, and 4 for 400GAUI-4.

Figure 120G–2—100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M insertion loss budget at 26.56 GHz

The nominal signaling rate for each lane is 53.125 GBd.

120G.1.1 Bit error ratio

The bit error ratio (BER) when processed by the PMA according to Clause 135 for 100GAUI-1 C2M or Clause 120 for 200GAUI-2 or 400GAUI-4 C2M shall be less than 10^{-5} .

120G.2 Compliance point definitions

The electrical characteristics for the 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M interfaces are defined at compliance points for the host and module. Reference test fixtures, called compliance boards, are used to access the electrical specification parameters.

Figure 120G–3 depicts the location of compliance points when measuring 100GAUI-1, 200GAUI-2, or 400GAUI-4 C2M host compliance. The output of the Host Compliance Board (HCB) is used to verify the host electrical output signal at TP1a. The input of the HCB at TP4a is used to verify the host input compliance.

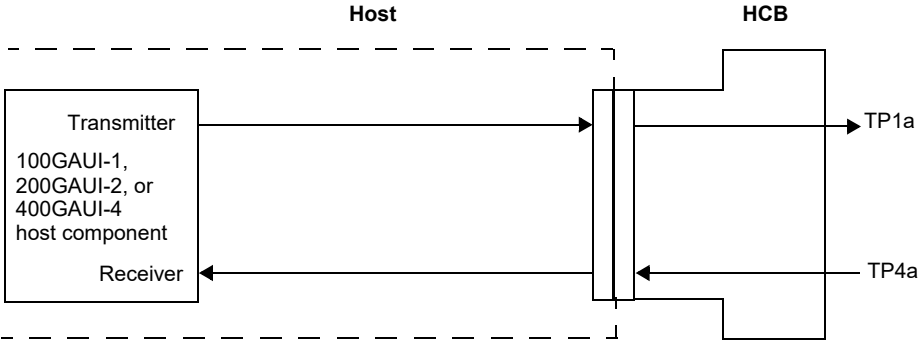


Figure 120G–3—Host compliance points

Figure 120G–4 depicts the location of compliance points when measuring 100GAUI-1, 200GAUI-2, or 400GAUI-4 C2M module compliance. The output of the Module Compliance Board (MCB) is used to verify the module electrical output signal at TP4. The input of the MCB at TP1 is used to verify the module input compliance.

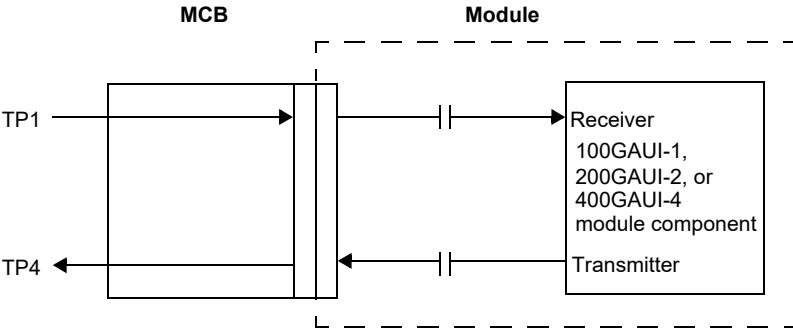


Figure 120G–4—Module compliance points

Additional details on the requirements for the MCB and HCB are given in 120G.5.4.

120G.3 Electrical characteristics

120G.3.1 Host output characteristics

The host output shall meet the specifications given in Table 120G–1.

A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for all output signal measurements, unless otherwise specified.

Table 120G–1—Host output characteristics at TP1a

Parameter	Reference	Value	Units
Signaling rate, each lane (range)		53.125 ± 50 ppm ^a	GBd
DC common-mode output voltage (max)	120G.5.1	2.8	V
DC common-mode output voltage (min)	120G.5.1	−0.3	V
Single-ended output voltage (max)	120G.5.1	3.3	V
Single-ended output voltage (min)	120G.5.1	−0.4	V
Peak-to-peak AC common-mode voltage (max) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	120G.5.1	32 80	mV
Differential peak-to-peak output voltage (max) Transmitter disabled Transmitter enabled	120G.5.1	35 750	mV
Steady-state voltage, v_f (max)	120G.5.3	375	mV
Eye height (min)	120G.3.1.5	10	mV
Vertical eye closure, VEC (max)	120G.3.1.5	12	dB
Common-mode to differential-mode return loss, RL_{dc} (min)	120G.3.1.1	Equation (120G–1)	dB
Effective return loss, ERL (min)	120G.3.1.2	7.3	dB
Differential termination mismatch (max)	120G.3.1.3	10	%
Transition time (min) Host is requesting short mode ^b Host is requesting long mode ^b	120G.3.1.4	10 15	ps ps

^a For a PMA in the same package as the PCS sublayer. In other cases, the signaling rate is derived from the signaling rate presented to the PMA input lanes (see Figure 135–3 and Figure 120–3) by the adjacent PMA or FEC sublayers.

^b Short and long modes are defined in 120G.3.2.1.

120G.3.1.1 Host output common-mode to differential-mode return loss

The host output common-mode to differential-mode return loss shall meet Equation (120G–1) as illustrated in Figure 120G–5.

$$RL_{dc}(f) \geq \begin{cases} 25 - 22(f/53.125) & 0.05 \leq f \leq 26.56 \\ 19 - 10(f/53.125) & 26.56 < f \leq 50 \end{cases} \quad (120G-1)$$

where

RL_{dc} is the common-mode to differential-mode return loss in dB
 f is the frequency in GHz

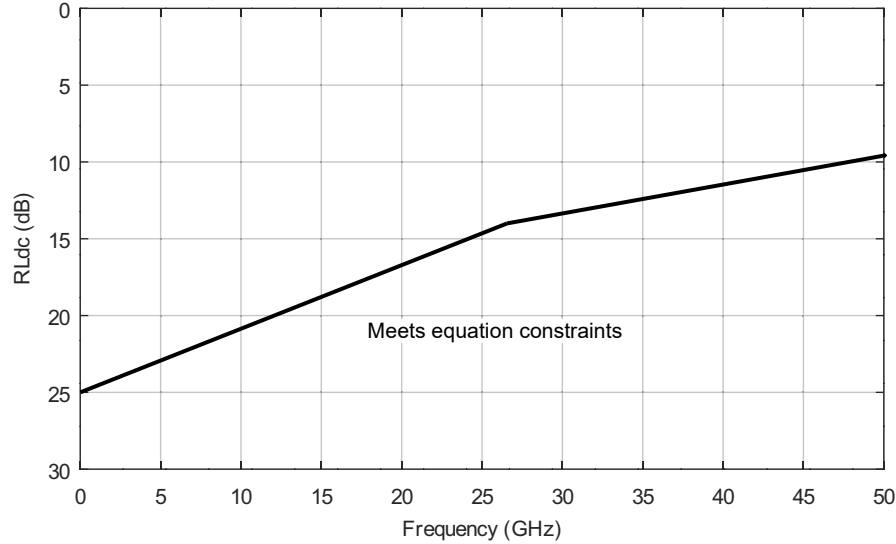


Figure 120G-5—Host output common-mode to differential-mode return loss

120G.3.1.2 Host output effective return loss (ERL)

The ERL of the host output at TP1a is computed using the procedure in 93A.5 with the values in Table 120G-2 and Table 120F-8, and with the value of T_{fx} equal to twice the delay between the test fixture test connector and the test fixture host-facing connection minus 0.2 ns.

Host output ERL at TP1a shall be greater than or equal to ERL (min) specified in Table 120G-1.

Table 120G-2—Host output and input ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	800	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—

120G.3.1.3 Differential termination mismatch

Differential termination mismatch is defined in 86A.5.3.2.

120G.3.1.4 Transition time

Transition time is defined in 120E.3.1.5, with the exception that the waveform is observed through a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 40 GHz.

120G.3.1.5 Host output eye height and vertical eye closure (VEC)

Figure 120G–6 depicts an example host output eye height and vertical eye closure (VEC) test configuration. Host output eye height and VEC are measured at TP1a using compliance boards defined in 120G.5.4. Eye height and VEC are measured according to the method described in 120G.5.2.

All counter-propagating signals are asynchronous to the co-propagating signals using the PRBS13Q²¹ (see 120.5.11.2.1) or PRBS31Q²² (see 120.5.11.2.2) pattern, or a valid 100GBASE-R, 200GBASE-R, or 400GBASE-R signal. For the case where PRBS13Q or PRBS31Q are used with a common clock, there is at least 31 UI delay between the patterns on one lane and any other lane, so that the symbols on each lane are not correlated. The crosstalk generator is calibrated at TP4 using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, with target differential peak-to-peak voltage of 845 mV and transition time (see 120G.3.1.4) of 8.5 ps.

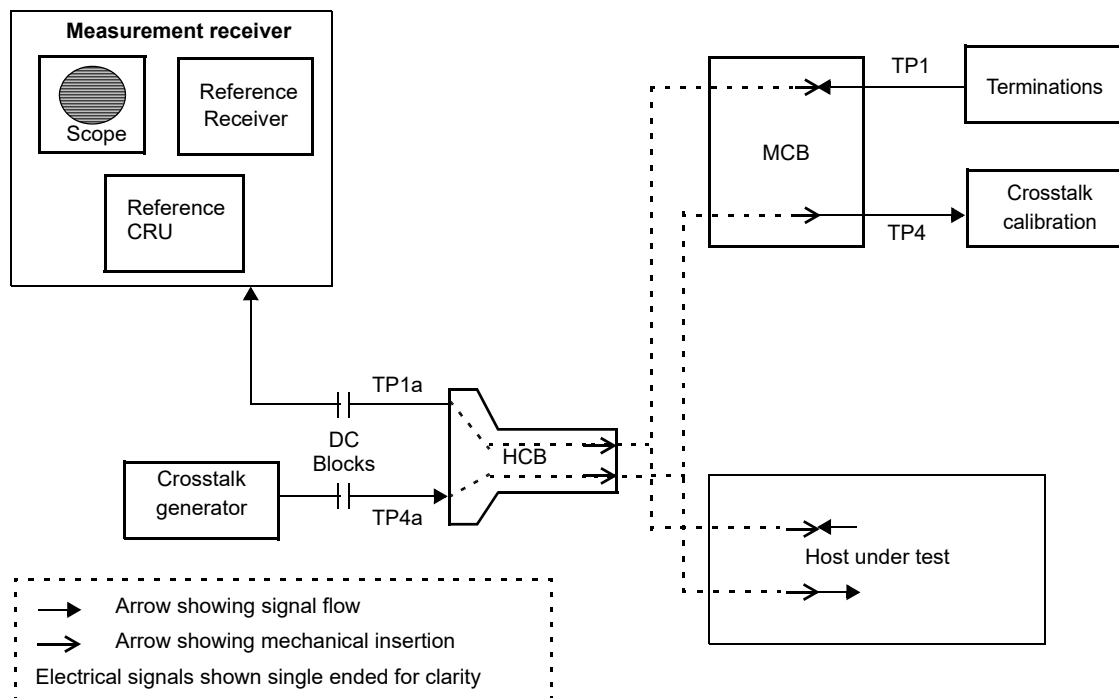


Figure 120G–6—Example host output test configuration

120G.3.2 Module output characteristics

The module output shall meet the specifications given in Table 120G–3. The low-frequency 3 dB cutoff of the output AC-coupling within the module should be less than 100 kHz. A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for all output signal measurements, unless otherwise specified.

²¹ PRBS13Q is also referred to as Pattern 4 for some PAM4 optical PMDs (e.g., see Table 121–9).

²² PRBS31Q is also referred to as Pattern 3 for some PAM4 optical PMDs (e.g., see Table 121–9).

Table 120G–3—Module output characteristics at TP4

Parameter	Reference	Value	Units
Signaling rate, each lane (nominal)		53.125 ^a	GBd
Peak-to-peak AC common-mode voltage (max) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	120G.5.1	32 80	mV
Differential peak-to-peak output voltage (max) Short mode Long mode	120G.5.1	600 845	mV mV
Eye height (min)	120G.3.2.2	15	mV
Vertical eye closure, VEC (max)	120G.3.2.2	12	dB
Common-mode to differential-mode return loss, RL_{dc} (min)	120G.3.1.1	Equation (120G–1)	dB
Effective return loss, ERL (min)	120G.3.2.3	8.5	dB
Differential termination mismatch (max)	120G.3.1.3	10	%
Transition time (min)	120G.3.1.4	8.5	ps
DC common-mode voltage tolerance (range) Upper limit Lower limit	120G.3.2.4	2.85 -0.35	V V

^a The signaling rate range is derived from the PMD receiver input.

120G.3.2.1 Module output modes

The module output shall support two modes: short and long. The means of controlling the module output mode is implementation dependent. For each output mode, the module shall meet the requirements for eye height (min) and VEC (max) in Table 120G–3 for both near-end and far-end measurements (see 120G.3.2.2.1).

One method for configuring the module output mode is to configure the module for a particular host electrical interface. The mapping of the combination of IEEE 802.3 interface type and module output mode to a host electrical interface may be determined from Table 120G–4.

Table 120G–4—Module output mode mapping

IEEE 802.3 interface type	Module output mode	Host electrical interface
100GAUI-1 C2M	short	100GAUI-1-S C2M
100GAUI-1 C2M	long	100GAUI-1-L C2M
200GAUI-2 C2M	short	200GAUI-2-S C2M
200GAUI-2 C2M	long	200GAUI-2-L C2M
400GAUI-4 C2M	short	400GAUI-4-S C2M
400GAUI-4 C2M	long	400GAUI-4-L C2M

120G.3.2.2 Module output eye height and VEC

Figure 120G–7 depicts an example module output eye height and VEC test configuration. Module output eye height and VEC are measured at TP4 using compliance boards defined in 120G.5.4. For each module output mode, eye height and VEC are measured according to the method described in 120G.3.2.2.1 using both the near-end and far-end host channels.

NOTE—Driving the module output using an internally generated clock, rather than the recovered clock from the PMD input, might create output signals that are more optimistic than full-link operation.

All counter-propagating signals are asynchronous to the co-propagating signals using the PRBS13Q²³ (see 120.5.11.2.1) or PRBS31Q²⁴ (120.5.11.2.2) pattern, or a valid 100GBASE-R, 200GBASE-R, or 400GBASE-R signal. For the case where PRBS13Q or PRBS31Q are used with a common clock, there is at least 31 UI delay between the patterns on one lane and any other lane, so that the symbols on each lane are not correlated. The crosstalk generator is calibrated at TP1a using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, with target differential peak-to-peak voltage of 750 mV and transition time (see 120G.3.1.4) of 10 ps for short mode and 15 ps for long mode.

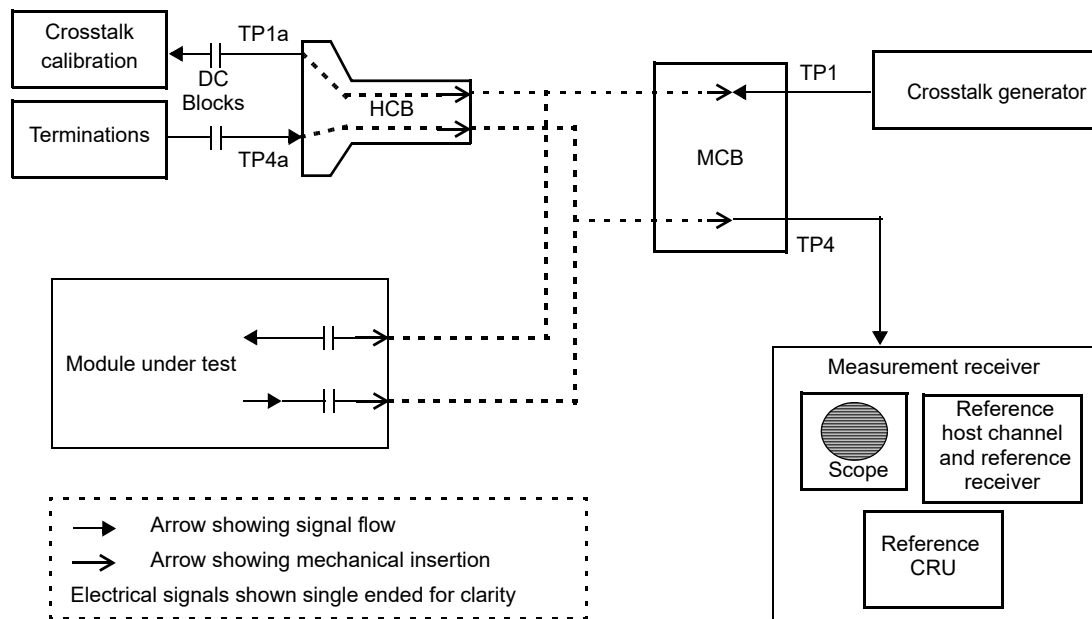


Figure 120G–7—Example module output test configuration

²³ PRBS13Q is also referred to as Pattern 4 for some PAM4 optical PMDs (e.g., see Table 121–9).

²⁴ PRBS31Q is also referred to as Pattern 3 for some PAM4 optical PMDs (e.g., see Table 121–9).

120G.3.2.2.1 Near-end and far-end eye measurement methodology

The signal measured at TP4 is first convolved with a reference host channel. The reference host channel is the host receiver printed circuit board (PCB) signal path $S^{(\text{HOSPR})}$ defined in 162.11.7.1.1 with the exceptions that the length z_p for each test is provided in Table 120G–5, and C_0 and C_1 are both 0 nF. The eye height and VEC are measured using the method in 120G.5.2.

Table 120G–5—PCB length for module output measurements

Module output mode	Host channel type	PCB length, z_p (mm)
Short	near-end	0
Short	far-end	133
Long	near-end	80
Long	far-end	244.7

120G.3.2.3 Module output ERL

The ERL of the module output at TP4 is computed using the procedure in 93A.5 with the values in Table 120G–6 and Table 120F–8, and with the value of T_{fx} equal to twice the delay between the test fixture test connector and the test fixture module-facing connection minus 0.2 ns.

Module output ERL at TP4 shall be greater than or equal to ERL (min) specified in Table 120G–3.

Table 120G–6—Module output and input ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	400	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—

120G.3.2.4 Module output common-mode voltage tolerance

DC common-mode voltage is generated by the host. A compliant module meets the output specifications with any DC common-mode voltage (see 120G.5.1), as measured at TP4, within the range specified in Table 120G–3.

NOTE—The specified voltages allow for the effects of ground offset voltage.

120G.3.3 Host input characteristics

The host input shall meet the specifications given in Table 120G–7. Channel equalization is provided by an adaptive equalizer in the host.

Table 120G–7—Host input characteristics

Parameter	Reference	Test point	Value	Units
Signaling rate, each lane (range)	120G.3.3.1	TP4a	53.125 ± 100 ppm	GBd
Differential peak-to-peak input voltage tolerance (min) for short mode for long mode	120G.5.1	TP4	600 845	mV
Peak-to-peak AC common-mode voltage tolerance (min) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	120G.5.1	TP4	32 80	mV
Differential-mode to common-mode return loss, RL_{cd} (min)	120G.3.3.3	TP4a	Equation (120G–2)	dB
Effective return loss, ERL (min)	120G.3.3.4	TP4a	7.3	dB
Host stressed input tolerance	120G.3.3.5	TP4	See 120G.3.3.5	
Differential termination mismatch (max)	120G.3.1.3	TP4a	10	%
Common-mode voltage ^a Min Max	120G.5.1	TP4a	–0.3 2.8	V

^a Generated by host, referred to host ground.

120G.3.3.1 Host input signaling rate

The host input shall comply with the requirements of 120G.3.3.5 for any signaling rate in the range specified in Table 120G–7.

120G.3.3.2 Host input AC common-mode voltage tolerance

A host input shall meet all other specifications with both $V_{CM_{LF}}$ and $V_{CM_{FB}}$ (see 120G.5.1) up to the limits specified in Table 120G–7.

120G.3.3.3 Host input differential-mode to common-mode return loss

The host input differential-mode to common-mode return loss shall meet Equation (120G–2) as illustrated in Figure 120G–8.

$$RL_{cd}(f) \geq \begin{cases} 22 - 16(f/53.125) & 0.05 \leq f \leq 26.56 \\ 19 - 10(f/53.125) & 26.56 < f \leq 50 \end{cases} \quad (120G-2)$$

where

RL_{cd} is the differential-mode to common-mode return loss in dB
 f is the frequency in GHz

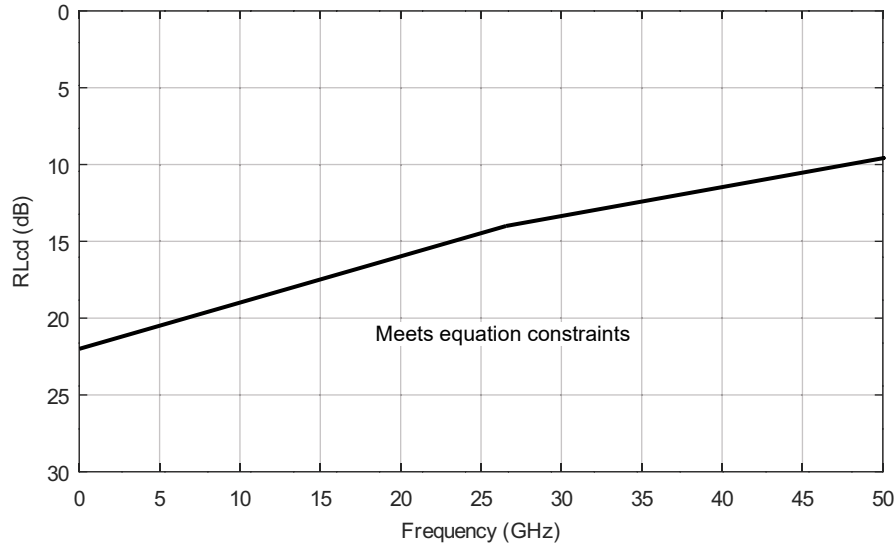


Figure 120G-8—Host input differential-mode to common-mode return loss

120G.3.3.4 Host input ERL

The ERL of the host input at TP4a is computed using the procedure in 93A.5 with the values in Table 120G-2 and Table 120F-8.

Host input ERL at TP4a shall be greater than or equal to ERL (min) specified in Table 120G-7.

120G.3.3.5 Host stressed input tolerance

Host stressed input tolerance is defined by the procedure described in 120G.3.3.5.1 through 120G.3.3.5.3.

The host under test shall meet the BER requirement in 120G.1.1:

- For either the short or long mode.
- With all sinusoidal jitter cases in Table 162-17.
- For any signaling rate in the range given in Table 120G-7.

120G.3.3.5.1 Host stressed input test setup

The host stressed input test setup is illustrated in Figure 120G-9. The stressed signal is applied at TP4a and is calibrated at TP4.

The stressed signal includes the following impairments:

- Sinusoidal jitter, random jitter, and bounded uncorrelated jitter.
- Counter-propagating crosstalk signals.

The pattern generator output equalization functional behavior is equivalent to the model shown in Figure 120F–3. The tap coefficients are not specified with the exception that the “no equalization” state is defined as setting coefficient $c(0)$ to 1 and all other coefficients to 0.

Bounded uncorrelated jitter may not be available in all stressed pattern generators or bit error ratio testers. It can be generated by driving the pattern generator external jitter modulation input with a low-pass filtered pseudo-random pattern. The pattern should be either PRBS7 or PRBS9 (see 83.5.10) with a signaling rate approximately 1/20 of the stressed pattern signaling rate (e.g., 2.656 GBd). The clock source for the PRBS generator is asynchronous to the pattern generator clock source. The low-pass filter has 20 dB/decade roll-off with a –3 dB corner frequency between 150 MHz and 300 MHz.

A crosstalk generator for counter-propagating signals representing the host output is used for calibration of the stress. The counter-propagating signals are asynchronous to the pattern generator. This crosstalk generator is not present in the test, being replaced by the output of the host under test.

The measurement receiver used for test calibration includes:

- A clock recovery unit (CRU) that acts as a high-pass jitter filter with a 3 dB corner frequency of 4 MHz and a slope of 20 dB/decade.
- A reference host channel to be configured for short or long mode far-end test calibration or long-mode near end verification as specified in 120G.3.2.2.1.
- A reference receiver as specified in 120G.5.2.

The test system meets the ERL specification given in 120G.3.2.3 when measured at TP4.

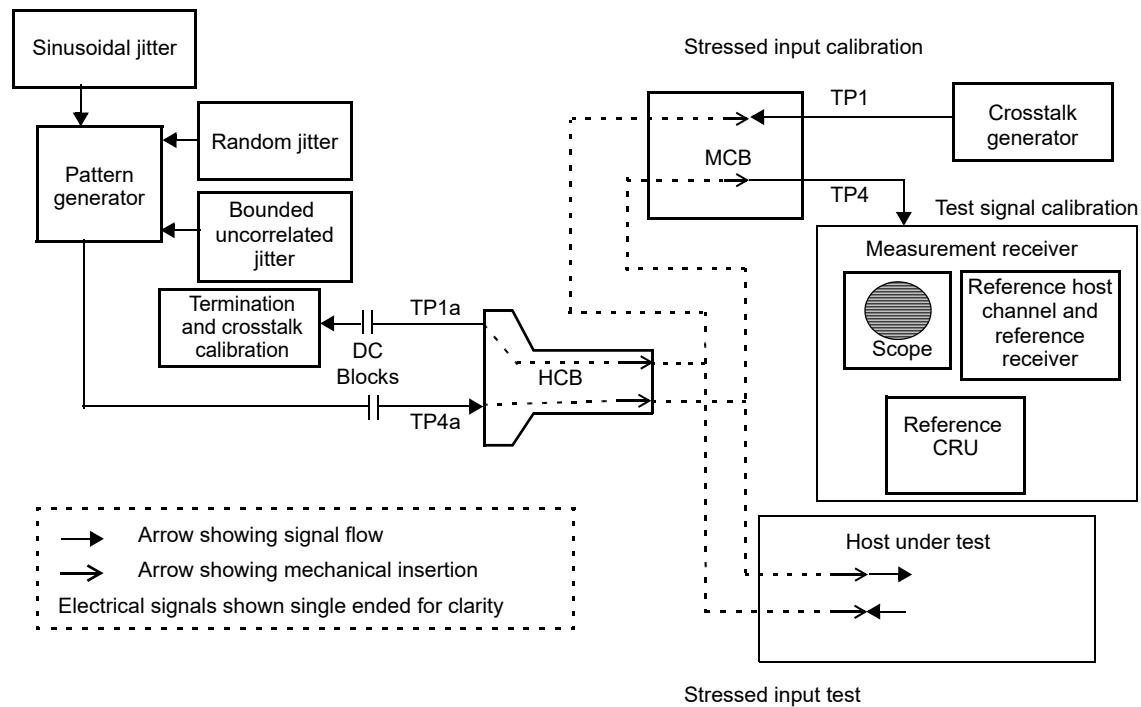


Figure 120G-9—Example host stressed input test

120G.3.3.5.2 Host stressed input test calibration

The host stressed input signal is calibrated by the following procedure.

- The pattern generator is set to generate a PRBS13Q pattern (see 120.5.11.2.1). The transition time (see 120G.3.1.4) measured at TP4a, with the pattern generator output equalization configured for “no equalization”, is as specified in Table 120G-8. The initial signal level is set to the differential peak-to-peak input voltage tolerance given in Table 120G-7.
- Sinusoidal jitter is applied with frequency and amplitude per case F in Table 162-17.
- Random jitter and bounded uncorrelated jitter are added to the sinusoidal jitter such that the jitter profile of the signal at TP4a approximates J_{RMS} (max) and J_{4u} (max) and complies with the even-odd jitter (max) specification in Table 120F-1. These are initial jitter values, that will be modified by the addition of crosstalk in step e) and adjustment of random jitter in step g).
- The HCB is plugged into the MCB.
- The counter-propagating crosstalk signals are calibrated at TP1a using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, to the differential peak-to-peak voltage and transition time (see 120G.3.1.4) specified in Table 120G-8. The crosstalk signal transition time and amplitude are calibrated with a PRBS13Q pattern. The crosstalk pattern is changed to PRBS31Q (see 120.5.11.2.2), scrambled idle (see 82.2.11 and 119.2.4.9), or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R signal for stressed signal calibration (see step g)). If the PRBS31Q pattern is used with a common clock, there is at least 31 UI delay between the PRBS31Q patterns on one lane and any other lane.
- The reference host channel is configured in the same way as in 120G.3.2.2.1 using the parameters in Table 120G-5 for host channel type and the requested module mode (short or long).

- g) Eye height and VEC are measured at TP4 as described in 120G.5.2. The pattern generator amplitude and random jitter are adjusted, while the pattern generator output equalization and reference receiver settings are adjusted to minimize far-end VEC, so that the far-end eye height of the smallest eye matches the target value, near-end VEC is within the limits in Table 120G–8, and differential peak-to-peak output voltage, near-end VEC, and eye height are within the limits in Table 120G–3. The differential peak-to-peak voltage measured at TP4 does not exceed the differential peak-to-peak input voltage tolerance given in Table 120G–7.

Table 120G–8—Host stressed input parameters

Parameter	Value	Units
Pattern generator transition time (target)	9	ps
Applied peak-to-peak sinusoidal jitter	Table 162–17	—
Eye height (target)	15	mV
Vertical eye closure, VEC (min)	12	dB
Vertical eye closure, VEC (max)	12.5	dB
Crosstalk differential peak-to-peak voltage	750	mV
Crosstalk transition time for short mode for long mode	10 15	ps

120G.3.3.5.3 Host stressed input test procedure

After the stress has been calibrated, the pattern generator is set to generate PRBS31Q, scrambled idle, or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R sequence. The HCB is unplugged from the MCB and is plugged into the host under test. The host electrical output is enabled on all lanes with any of the patterns above. The test is repeated with sinusoidal jitter set to each of the six cases in Table 162–17.

If the test is performed with PRBS31Q, the BER for the AUI under test may be calculated using the bit error counter in the PMA test pattern checker (see 120.5.11.2.2) as the number of bit errors divided by the number of received bits. The BER for the AUI under test is the average of the BER of each of its lanes.

If the test is performed with scrambled idle or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R sequence, the BER for the AUI under test may be calculated using the FEC decoder error counters (see 91.6 and 119.3.1), as the number of FEC symbol errors divided by the number of received bits.

The number of received bits may be estimated based on the test time.

Methods of extracting the received bit pattern and counting errors other than the ones described above may be used if they generate equivalent results.

120G.3.4 Module input characteristics

The module input shall meet the specifications given in Table 120G–9. Channel equalization is provided by an adaptive equalizer in the module.

Table 120G–9—Module input characteristics

Parameter	Reference	Test point	Value	Units
Signaling rate, each lane (range)	120G.3.4.1	TP1	53.125 ± 100 ppm	GBd
Differential pk-pk voltage tolerance (min)	120G.5.1	TP1a	750	mV
Peak-to-peak AC common-mode voltage tolerance (min) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	120G.5.1	TP1a	32 80	mV
Differential-mode to common-mode return loss, RL_{cd} (min)	120G.3.3.3	TP1	Equation (120G–2)	dB
Effective return loss, ERL (min)	120G.3.4.4	TP1	8.5	dB
Differential termination mismatch (max)	120G.3.1.3	TP1	10	%
Module stressed input tolerance	120G.3.4.3	TP1a	See 120G.3.4.3	
Single-ended voltage tolerance range (min)	120G.5.1	TP1a	–0.4 to 3.3	V
DC common-mode voltage tolerance (range) Upper limit Lower limit	120G.3.4.5	TP1	2.85 –0.35	V V

120G.3.4.1 Module input signaling rate

The module input shall comply with the requirements of 120G.3.4.3 for any signaling rate in the range specified in Table 120G–9.

120G.3.4.2 Module input AC common-mode voltage tolerance

A module input shall meet all other specifications with both $V_{CM_{LF}}$ and $V_{CM_{FB}}$ (see 120G.5.1) up to the limits specified in Table 120G–9.

120G.3.4.3 Module stressed input tolerance

Module stressed input tolerance is defined by the procedure described in 120G.3.4.3.1 through 120G.3.4.3.3.

The module under test shall meet the BER requirement in 120G.1.1:

- For both the high-loss and low-loss cases (see 120G.3.4.3.2), calibrated separately.
- With all sinusoidal jitter cases in Table 162–17.
- For any signaling rate in the range given in Table 120G–9.

120G.3.4.3.1 Module stressed input test setup

The module stressed input test setup is illustrated in Figure 120G–10. The stressed signal is applied at TP1 and is calibrated at TP1a.

The stressed signal includes the following impairments:

- Sinusoidal jitter, random jitter, and bounded uncorrelated jitter.
- Frequency-dependent attenuation representing the host channel, which may be implemented with PCB traces. The frequency-dependent attenuation is used only for the high-loss case (see 120G.3.4.3.2).
- Counter-propagating crosstalk signals.

The pattern generator output equalization is described in 120G.3.3.5.1.

Bounded uncorrelated jitter may not be available in all stressed pattern generators or bit error ratio testers. It can be generated as described in 120G.3.3.5.2.

A crosstalk generator for counter-propagating signals representing the module output is used for calibration of the stress. The counter-propagating signals are asynchronous to the pattern generator. This crosstalk generator is not present in the test, being replaced by the output of the module under test.

The measurement receiver used for test calibration includes:

- A clock recovery unit (CRU) that acts as a high-pass jitter filter with a 3 dB corner frequency of 4 MHz and a slope of 20 dB/decade.
- A reference receiver as specified in 120G.5.2.

The test system meets the ERL specification given in 120G.3.1.2 when measured at TP1a.

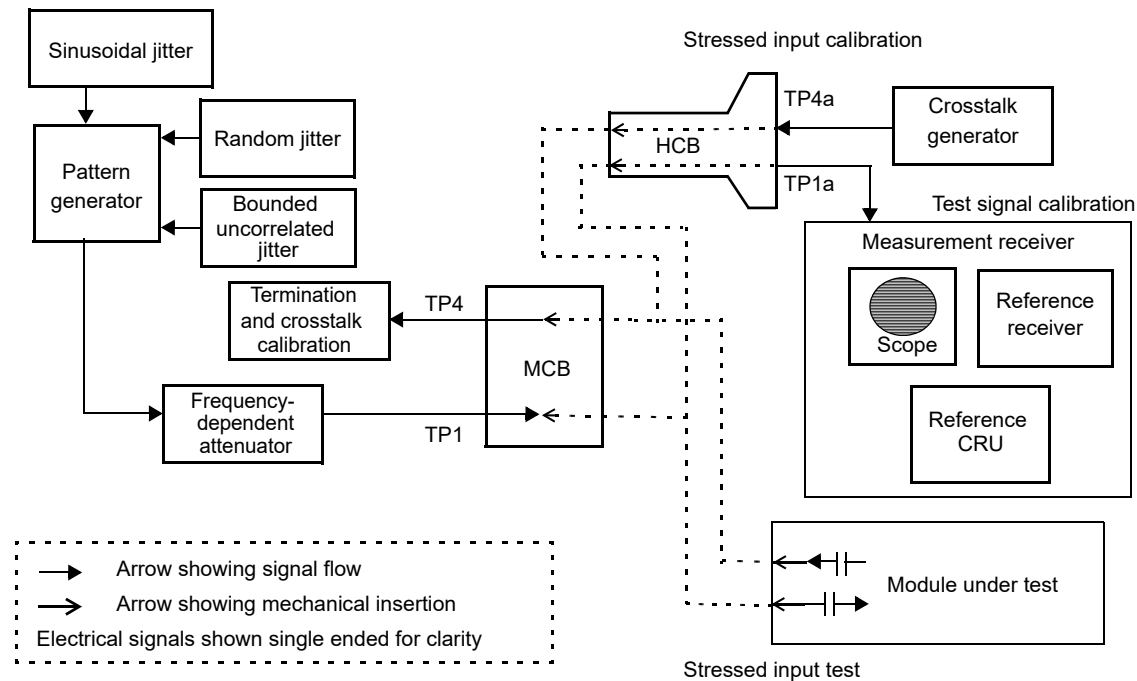


Figure 120G-10—Example module stressed input test

120G.3.4.3.2 Module stressed input test calibration

The stressed input signal is calibrated by the following procedure.

- The pattern generator is set to generate a PRBS13Q pattern (see 120.5.11.2.1). The transition time (see 120G.3.1.4) measured at the output of the pattern generator, with the pattern generator output equalization configured for “no equalization” (see 120G.3.3.5.1), is as specified in Table 120G-10. The initial signal level is set to the differential peak-to-peak input voltage tolerance given in Table 120G-9.
- Sinusoidal jitter is applied with frequency and amplitude per case F in Table 162-17.
- Random jitter and bounded uncorrelated jitter are added to the sinusoidal jitter such that the jitter profile of the signal at the output of the pattern generator approximates $J_{RMS}(\max)$ and $J_{4u}(\max)$ and complies with the even-odd jitter (max) specification in Table 120F-1. These are initial jitter values, that will be modified by the addition of crosstalk in step e) and adjustment of random jitter in step g).
- The HCB is plugged into the MCB.
- The counter-propagating crosstalk signals are calibrated at TP4 using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, to the differential peak-to-peak voltage and transition time (see 120G.3.1.4) specified in Table 120G-10. The crosstalk signal transition time and amplitude are calibrated with a PRBS13Q pattern. The crosstalk pattern is changed to PRBS31Q (see 120.5.11.2.2), scrambled idle (see 82.2.11, 119.2.4.9), or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R signal for stressed signal calibration (see step g)). If the PRBS31Q pattern is used with a common clock, there is at least 31 UI delay between the PRBS31Q patterns on one lane and any other lane.
- For the low-loss signal calibration, the output of the pattern generator is fed directly to the MCB input (TP1). For the high-loss signal calibration, the frequency-dependent attenuator is configured such that the insertion loss of the test channel from the output of the pattern generator to TP1a

approximates the target insertion loss, $ILDD_{channel}$, given by Equation (120G–3). The target insertion loss is 18.2 dB at 26.56 GHz, representing 16 dB channel loss with an additional 2.2 dB, which provides some allowance for the difference between the effective insertion loss of the pattern generator output and the worst-case host transmitter package loss. To achieve the target test channel insertion loss, the scattering parameters of the frequency-dependent attenuator should approximate those for a PCB transmission line calculated from Equation (93A–13) and Equation (93A–14) using $z_p = 296$ mm in length and the relevant parameter values given in Table 162–21. Additional frequency-independent delay may be included in the test channel. The magnitude of the insertion loss of the target frequency-dependent attenuator scattering parameters, $ILDD_{FDA}$, is represented by Equation (120G–4). $ILDD_{channel}$ and $ILDD_{FDA}$ are illustrated in Figure 120G–11.

- g) Eye height and VEC are measured at TP1a as described in 120G.5.2. For the high-loss case, an exception is made that the reference receiver CTLE is limited to settings where $g_{DC} + g_{DC2}$ is less than or equal to -10.5 dB.
- h) The pattern generator amplitude, output equalization, and random jitter are adjusted together to minimize VEC, so that the eye height of the smallest eye matches the target value, VEC is within the limits in Table 120G–10, and the differential peak-to-peak voltage measured at TP1a does not exceed the differential peak-to-peak input voltage tolerance given in Table 120G–9. The pattern generator output equalization has to be set such that for the resulting signal, the same VEC is achieved with or without the limitations on g_{DC} and g_{DC2} in item g).

Other calibration procedures resulting in a signal that meets these requirements may be used.

Table 120G–10—Module stressed input parameters

Parameter	Value	Value
Pattern generator transition time (target)	9	ps
Applied peak-to-peak sinusoidal jitter	Table 162–17	—
Eye height (target)	10	mV
Vertical eye closure, VEC (min)	12	dB
Vertical eye closure, VEC (max)	12.5	dB
Crosstalk differential peak-to-peak voltage	845	mV
Crosstalk transition time	8.5	ps

$$ILdd_{channel}(f) = 1.7962\sqrt{f} + 0.2463f + 0.003405f^2 \quad (120G-3)$$

where

$ILdd_{channel}(f)$ is the target pattern generator output to TP1a insertion loss in dB
 f is the frequency in GHz

$$ILdd_{FDA}(f) = 0.981 \sqrt{f} + 0.2463f \quad (120G-4)$$

where

$ILdd_{FDA}(f)$ is the target high-loss frequency-dependent attenuator insertion loss in dB
 f is the frequency in GHz

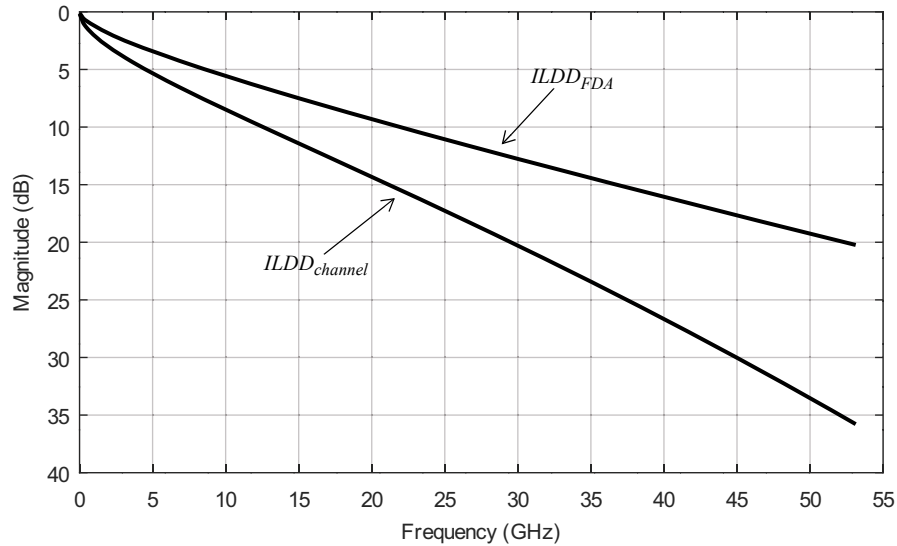


Figure 120G-11—Module stressed input insertion loss targets

120G.3.4.3.3 Module stressed input test procedure

After the stress has been calibrated, the pattern generator is set to generate PRBS31Q, scrambled idle, or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R sequence. The HCB is unplugged from the MCB and the module under test is plugged into the MCB. The module electrical output is enabled on all lanes with any of the patterns above. The test is repeated with sinusoidal jitter set to each of the six cases in Table 162-17.

If the test is performed with PRBS31Q, the BER for the AUI under test may be calculated using the bit error counter in the PMA test pattern checker (see 120.5.11.2.2) as the number of bit errors divided by the number of received bits. The BER for the AUI under test is the average of the BER of each of its lanes.

If the test is performed with scrambled idle or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R sequence, the BER for the AUI under test may be calculated by placing the module under test into local loopback (see 120.5.9) and feeding the module output into a compliant host or its equivalent. The BER for the AUI under test is calculated using the FEC decoder error counters (see 91.6 and 119.3.1), as the number of FEC symbol errors divided by the number of received bits.

The number of received bits may be estimated based on the test time.

Methods of extracting the received bit pattern and counting errors other than the ones described above may be used if they generate equivalent results.

120G.3.4.4 Module input ERL

The ERL of the module input at TP1 is computed using the procedure in 93A.5 with the values in Table 120G–6 and Table 120F–8.

Module input ERL at TP1 shall be greater than or equal to ERL (min) specified in Table 120G–9.

120G.3.4.5 Module input common-mode voltage tolerance

DC common-mode voltage is generated by the host. A compliant module meets the input specifications with any DC common-mode voltage (see 120G.5.1), as measured at TP1, within the range specified in Table 120G–9.

NOTE—The specified voltages allow for the effects of ground offset voltage.

120G.4 Channel insertion loss (recommended)

For worst-case package loss in the host, the insertion loss of the channel between the host and module C2M components (see Figure 120G–2) is recommended to meet Equation (120G–5), which is illustrated in Figure 120G–12. Note that for this equation the insertion loss at the Nyquist frequency is less than or equal to 16 dB. The insertion loss budget is illustrated in Figure 120G–2.

$$IL_{dd}(f) \leq \begin{cases} 0.05 + 1.8\sqrt{f} + 0.2513f & 0.01 \leq f \leq 26.56 \\ -12.4181 + 1.07f & 26.56 < f \leq 40 \end{cases} \quad (120G-5)$$

where

$IL_{dd}(f)$ is the channel insertion loss in dB
 f is the frequency in GHz

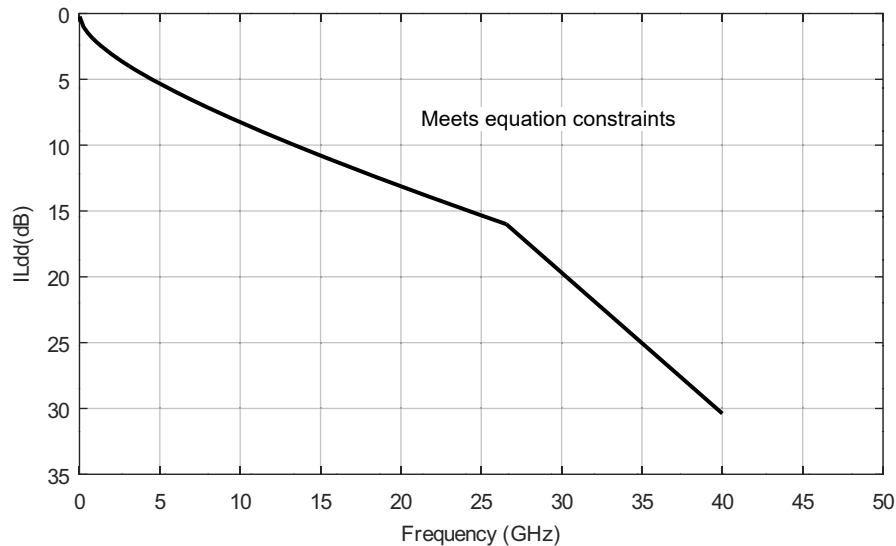


Figure 120G–12—Recommended channel insertion loss

120G.5 Measurement methodology

This subclause describes common measurement tools and methodology.

120G.5.1 Signal levels

The signal levels are as defined in 120E.3.1.2.

Low-frequency and full-band peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ and $V_{CM_{FB}}$, respectively, are defined by the method specified in 162.9.4.4 with the following exceptions:

- The peak-to-peak AC common-mode voltage is defined as the AC common-mode voltage range measured at TP1a or TP4 that includes all but 10^{-5} of the measured distribution, from 0.000005 to 0.999995 of the cumulative distribution.
- The condition for transmitter equalization to be turned off does not apply.

120G.5.2 Eye opening measurement method

The eye opening parameters eye height and VEC are measured with the effect of a reference receiver that includes receiver input referred noise, a continuous-time filter as defined in 93A.1.4.3, a receiver noise filter as defined in 93A.1.4.1, and a decision-feedback equalizer as defined in 93A.1.6, using the parameters specified in Table 120G–11. All parameters in Table 120G–11 apply to TP1a, TP4 far-end, and TP4 near-end unless indicated otherwise. The pattern used for output eye diagram measurements is PRBS13Q (see 120.5.11.2.1). Eye height and VEC parameters are defined by the following procedure.

Table 120G–11—Eye opening reference receiver parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	53.125	GBd
Receiver 3 dB bandwidth	f_r	$0.75 \times f_b$	GHz
Continuous time filter, DC gain for TP1a Range for $g_{DC2} = 0$ Range for $-1 \leq g_{DC2} < 0$ Range for $-2 \leq g_{DC2} < -1$ Range for $-3 \leq g_{DC2} < -2$ Step size	g_{DC}	 –2 to –9 –2 to –11 –4 to –10 –4 to –9 1.0	 dB
Continuous time filter, DC gain 2 for TP1a Range Step size	g_{DC2}	 –3 to 0 0.5	 dB
Continuous time filter, DC gain for TP4 near-end Range Step size	g_{DC}	 –5 to –1 1.0	 dB
Continuous time filter, DC gain 2 for TP4 near-end Range Step size	g_{DC2}	 –2 to 0 0.5	 dB
Continuous time filter, DC gain for TP4 far-end Range Step size	g_{DC}	 –9 to –2 1.0	 dB

Table 120G–11—Eye opening reference receiver parameter values (*continued*)

Parameter	Symbol	Value	Units
Continuous time filter, DC gain 2 for TP4 far-end Range Step size	g_{DC2}	–3 to –1 0.5	dB
Continuous time filter, zero frequency for $g_{DC} = 0$	f_z	12.58	GHz
Continuous time filter, pole frequencies	f_{p1} f_{p2}	20 28	GHz GHz
Continuous time filter, low-frequency pole/zero	f_{LF}	$f_b / 40$	GHz
Decision feedback equalizer (DFE) length	N_b	4	UI
Normalized DFE coefficient maximum limit $n = 1$ $n = 2$ $n = 3$ or 4	$bb_{\max}(n)$	0.4 0.15 0.1	— — —
Normalized DFE coefficient minimum limit $n = 1$ $n = 2$ $n = 3$ or 4	$bb_{\min}(n)$	0.1 –0.15 –0.05	— — —
One-sided noise spectral density	η_0	4.1×10^{-8}	V ² /GHz

Perform the following step once:

- a) Capture the PRBS13Q signal $y_1(k)$ with the effect of low-pass response equivalent to the specified receiver noise filter with associated parameter f_r in Table 120G–11 (instead of the test system response specified in 120G.3.1), and using a clock recovery unit with a corner frequency of 4 MHz and slope of 20 dB/decade. The signal is captured such that samples, after any post-processing as necessary, are evenly distributed throughout the measurement window and are numerous enough for an accurate normalized cumulative distribution function (CDF) to a probability of 10^{-5} without extrapolation.

Perform the following steps for each valid combination of g_{DC} and g_{DC2} as specified in Table 120G–11:

- b) Compute the response $y_2(k)$ by applying the effect of the continuous time filter to $y_1(k)$ using the associated parameters in Table 120G–11.
- c) Compute the linear fit pulse response $p_2(k)$ using the method defined in 162.9.4.1.1 with parameter M the same as for step a), D_p equal to 3, and N_p equal to 200.
- d) Compute the DFE sampling phase t_s and tap weights $b(n)$ for $p_2(k)$ according to the methodology in 93A.1.6 using the associated parameters in Table 120G–11.
- e) Compute the receiver input signal $y_{rx}(k)$ by applying the effect of the DFE to $y_2(k)$ using the sampling phase t_s and tap weights $b(n)$ determined in the previous step.
- f) Compute the variance of the noise at the output of the receive equalizer σ_N^2 based on the one-sided spectral density η_0 , provided in Table 120G–11, referred to the receiver noise filter input per Equation (93A–35).
- g) Compute an eye diagram from $y_{rx}(k)$, including the effect of Gaussian noise with variance calculated in the previous step, but taking into account that some noise from the measurement instrument is already in $y_{rx}(k)$.
- h) Compute Vmid, Vupp, Vlow, VCmid, VCupp, VClow from the eye diagram using 120E.4.2 steps 4) through 6) with the exception that the CDF of the signal voltage is calculated from the set of voltage samples within the time interval $t_s \pm 0.05$ UI and with accumulated probability for each sample

weighted by the function $w(t)$ defined by Equation (120G–6). As an example, $UPPCDF1$ may be calculated using Equation (120G–8) from a set of N samples $\{v_i, t_i\}$.

- i) The eye height is the minimum of V_{mid} , V_{upp} , and V_{low} .
- j) Compute VEC using Equation (120E–4) and the values computed in step h) with the exception that $VM3$, $VM2$, $VM1$, and $VM0$ are the average values over the time interval $t_s \pm 0.05$ UI instead of “within 0.025 UI of time TC_{mid} ”.

The values of eye height and VEC are the values obtained with the combination of g_{DC} and g_{DC2} that produces the minimum value of VEC where eye height also complies with the specification for eye height (min) as specified for the interface.

$$w(t) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma_r} \exp\left(-\frac{1}{2}\left(\frac{x(t)}{\sigma_r}\right)^2\right) & |x(t)| \leq 0.05 \\ 0 & |x(t)| > 0.05 \end{cases} \quad (120G-6)$$

$$x(t) = (t - t_s)f_b \quad (120G-7)$$

$$UPPCDF1(v) = \frac{1}{A} \sum_{i=1}^N w(t_i) u(v_i, VC_{upp}, v) \quad (120G-8)$$

$$u(x, a, b) = \begin{cases} 1 & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases} \quad (120G-9)$$

$$A = \sum_{i=1}^N w(t_i) \quad (120G-10)$$

where:

$w(t)$	is the weighting function
$x(t)$	is the deviation from the sampling time t_s in UI
σ_r	is the standard deviation in UI and is equal to 0.02 UI
f_b	is the signaling rate in Hz
t	is time in seconds
A	is a factor to normalize the weighting function over the sample set

120G.5.3 Steady-state voltage

The steady-state voltage v_f is defined as the sum of the linear fit pulse $p(1)$ through $p(M \times N_p)$ divided by M with the specific equalization used by the transmitter. N_v is set equal to N_p . The linear fit procedure for obtaining $p(k)$ and the values of M and N_p are defined in 162.9.4.1.1.

120G.5.4 HCB/MCB characteristics

HCB characteristics are described in 162B.2 where the HCB performs the equivalent function as the TP2 or TP3 test fixture. The MCB characteristics are described in 162B.3 where the MCB performs the equivalent

functionality as the cable assembly test fixture. The mated compliance board characteristics are described in 162B.4.

120G.6 Protocol implementation conformance statement (PICS) proforma for Annex 120G, Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)²⁵

120G.6.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 120G, Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

120G.6.2 Identification

120G.6.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

120G.6.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Annex 120G, Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
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²⁵Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

120G.6.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
NOL	Number of differential AC-coupled lanes	120G.1	One, two, or four independent data paths in each direction for 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M, respectively	M	Yes []
BER	Meets BER requirement	120G.1.1	See 120G.1.1	M	Yes []
*PCSPKG	PMA in same package as PCS.			O	Yes [] No []

120G.6.4 PICS proforma tables for Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)

120G.6.4.1 Host output

Item	Feature	Subclause	Value/Comment	Status	Support
TH1	Signaling rate, each lane, PMA in same package as PCS	120G.3.1	Per Table 120G–1	PCSPKG:M	Yes [] N/A []
TH2	Signaling rate, each lane, PMA not in same package as PCS	120G.3.1	Derived from PMA input lanes	!PCSPKG:M	Yes [] N/A []
TH3	DC common-mode output voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH4	Single-ended output voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH5	Low-frequency peak-to-peak AC common-mode voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH6	Full-band peak-to-peak AC common-mode voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH7	Differential peak-to-peak voltage, transmitter disabled	120G.3.1	Per Table 120G–1	M	Yes []
TH8	Differential peak-to-peak voltage, transmitter enabled	120G.3.1	Per Table 120G–1	M	Yes []
TH9	Steady-state voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH10	Eye height	120G.3.1	Per Table 120G–1	M	Yes []
TH11	VEC	120G.3.1	Per Table 120G–1	M	Yes []
TH12	Common-mode to differential-mode return loss	120G.3.1.1	Per Table 120G–1	M	Yes []
TH13	ERL	120G.3.1.2	Per Table 120G–1	M	Yes []
TH14	Differential termination mismatch	120G.3.1.3	Per Table 120G–1	M	Yes []
TH15	Transition time	120G.3.1.4	Per Table 120G–1	M	Yes []

120G.6.4.2 Module output

Item	Feature	Subclause	Value/Comment	Status	Support
TM1	Signal rate, each lane, nominal	120G.3.2	Per Table 120G–3	M	Yes []
TM2	Signal rate, each lane, range	120G.3.2	Derived from PMD receiver input.	M	Yes []
TM3	Low-frequency peak-to-peak AC common-mode voltage	120G.3.2	Per Table 120G–3	M	Yes []
TM4	Full-band peak-to-peak AC common-mode voltage	120G.3.2	Per Table 120G–3	M	Yes []
TM5	Peak-to-peak differential output voltage	120G.3.2	Per Table 120G–3	M	Yes []
TM6	Near-end eye height	120G.3.2	Per Table 120G–3	M	Yes []
TM7	Near-end VEC	120G.3.2	Per Table 120G–3	M	Yes []
TM8	Far-end eye height	120G.3.2	Per Table 120G–3	M	Yes []
TM9	Far-end VEC	120G.3.2	Per Table 120G–3	M	Yes []
TM10	Common-mode to differential-mode return loss	120G.3.2	Per Table 120G–3	M	Yes []
TM11	ERL	120G.3.2	Per Table 120G–3	M	Yes []
TM12	Differential termination mismatch	120G.3.2	Per Table 120G–3	M	Yes []
TM13	Transition time	120G.3.2	Per Table 120G–3	M	Yes []
TM14	DC common-mode voltage	120G.3.2	Per Table 120G–3	M	Yes []

120G.6.4.3 Host input

Item	Feature	Subclause	Value/Comment	Status	Support
RH1	Adaptive equalizer in host	120G.3.3		M	Yes []
RH2	Signaling rate tolerance, each lane	120G.3.3	Per Table 120G–7	M	Yes []
RH3	Differential peak-to-peak input voltage tolerance	120G.3.3	Per Table 120G–7	M	Yes []
RH4	Common-mode to differential-mode return loss	120G.3.3	Per Table 120G–7	M	Yes []
RH5	ERL	120G.3.3	Per Table 120G–7	M	Yes []
RH6	Differential termination mismatch	120G.3.3	Per Table 120G–7	M	Yes []
RH7	Common-mode voltage	120G.3.3	Per Table 120G–7	M	Yes []
RH8	Stressed input test	120G.3.3.5		M	Yes []

120G.6.4.4 Module input

Item	Feature	Subclause	Value/Comment	Status	Support
RH1	Adaptive equalizer in module	120G.3.4		M	Yes []
RH2	Signaling rate tolerance, each lane	120G.3.4	Per Table 120G–9	M	Yes []
RH3	Differential peak-to-peak input voltage tolerance	120G.3.4	Per Table 120G–9	M	Yes []
RH4	Common-mode to differential-mode return loss	120G.3.4	Per Table 120G–9	M	Yes []
RH5	ERL	120G.3.4	Per Table 120G–9	M	Yes []
RH6	Differential termination mismatch	120G.3.4	Per Table 120G–9	M	Yes []
RH7	Single-ended voltage tolerance	120G.3.4	Per Table 120G–9		
RH8	Common-mode voltage tolerance	120G.3.4	Per Table 120G–9	M	Yes []
RH9	Stressed input test	120G.3.4.3		M	Yes []

Annex 135A

(informative)

50 Gb/s and 100 Gb/s PMA sublayer partitioning examples

135A.2 Partitioning examples of 100GBASE-P PHYs

Replace Figure 135A–6 with the following figure:

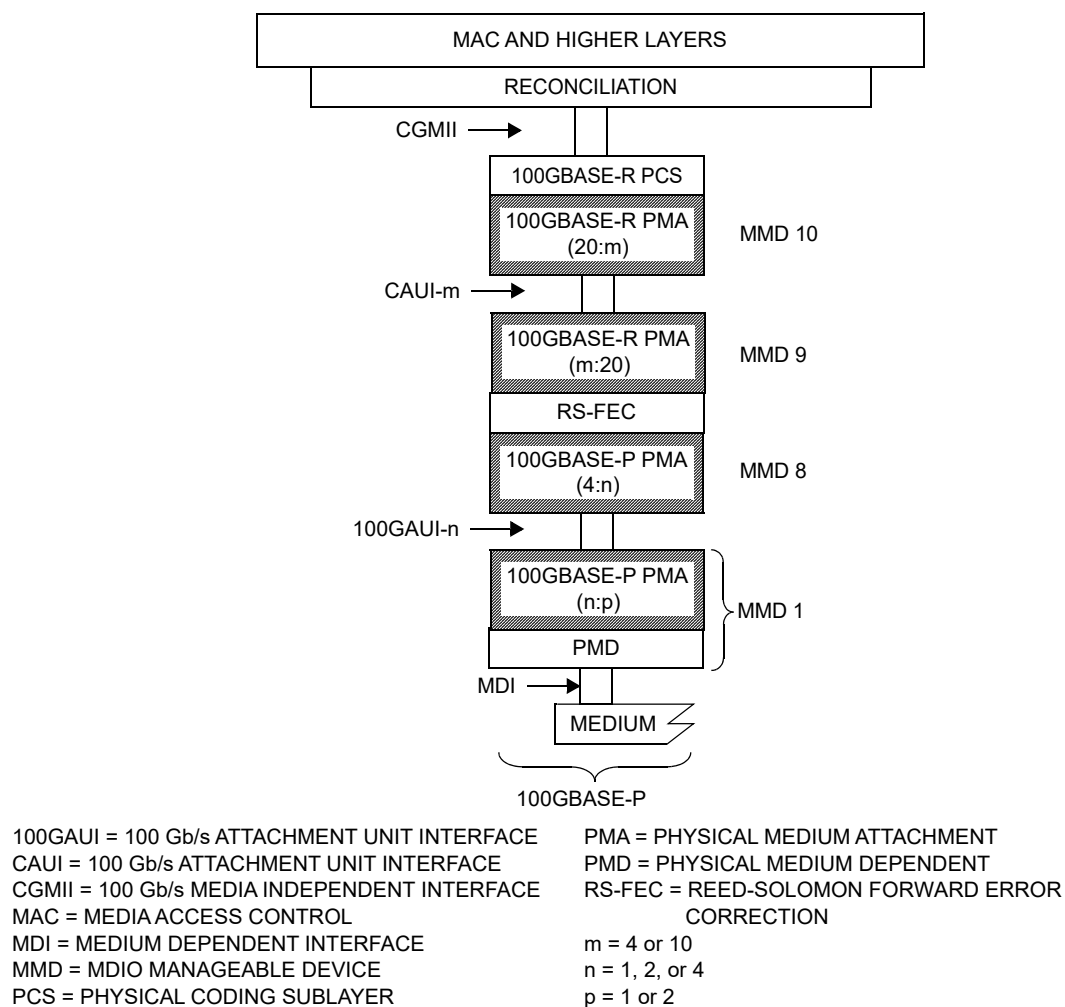
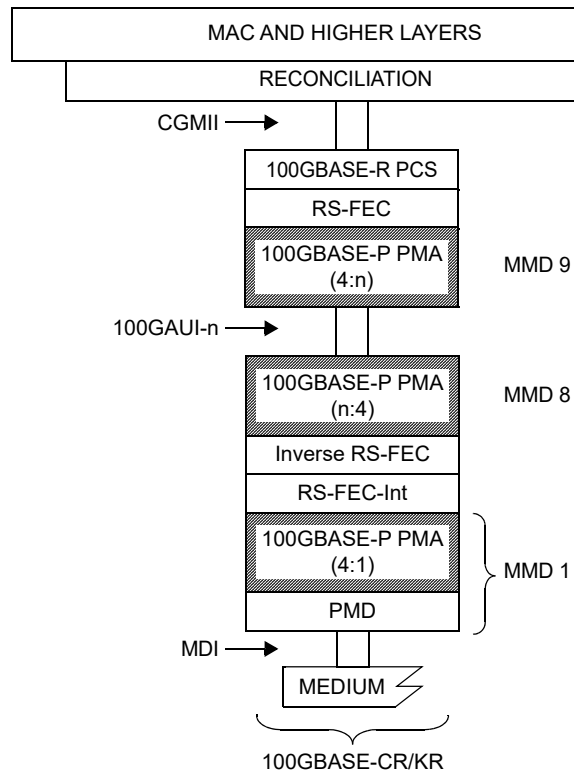


Figure 135A–6—Example 100GBASE-P PHY with CAUI-n and 100GAUI-n

Insert a new paragraph and Figure 135A–7a after Figure 135A–7 in 135A.3.1 as follows:

Figure 135A–7a depicts an example of a 100GBASE-CR/KR PHY with 100GAUI-n interface and Inverse RS-FEC.



100GAUI = 100 Gb/s ATTACHMENT UNIT INTERFACE
CGMII = 100 Gb/s MEDIA INDEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
MMD = MDIO MANAGEABLE DEVICE
PCS = PHYSICAL CODING SUBLAYER
PMA = PHYSICAL MEDIUM ATTACHMENT

PMD = PHYSICAL MEDIUM DEPENDENT
RS-FEC = REED-SOLOMON FORWARD ERROR CORRECTION
RS-FEC-Int = REED-SOLOMON FORWARD ERROR CORRECTION CODEWORD-INTERLEAVED
n = 1, 2, or 4

Figure 135A–7a—Example 100GBASE-CR/KR PHY with RS-FEC and RS-FEC-Int

Insert new annexes Annex 162A through Annex 162D corresponding to Clause 162 as follows:

Annex 162A

(informative)

Transmitter, receiver and channel parameters associated with test points TP0 and TP5 for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

162A.1 Overview

This annex provides information on transmitter, receiver, and channel parameters associated with test points TP0 and TP5, which might not be testable in an implemented system. TP0 and TP5 test points are illustrated in the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 link block diagram of Figure 162–2. This annex also provides information on channel characteristics.

162A.2 Transmitter characteristics at TP0

The recommended transmitter characteristics at TP0 as measured at TP0v are described in 163.9.2.

162A.3 Receiver characteristics at TP5

The recommended receiver characteristics at TP5 as measured at TP5v are described in 163.9.3.

162A.4 Transmitter and receiver differential printed circuit board (PCB) trace insertion loss

The recommended maximum and minimum insertion loss allocation for the transmitter or receiver differential controlled impedance PCBs are determined using Equation (162A–1) and Equation (162A–2), respectively, and illustrated in Figure 162A–1. Note that the recommended maximum insertion loss allocation for the transmitter or receiver differential controlled impedance PCBs with allowances for ball grid array (BGA) footprint and host connector footprints is 6.875 dB at 26.56 GHz and the recommended minimum insertion loss allocation for the transmitter or receiver differential controlled impedance PCBs is 2.3 dB at 26.56 GHz.

$$ILdd_{PCB}(f) \leq ILdd_{PCBmax}(f) = 0.9809(0.471\sqrt{f} + 0.1194f + 0.002f^2) \quad (162A-1)$$

for 0.01 GHz $\leq f \leq$ 50 GHz

where

$ILdd_{PCB}(f)$	is the insertion loss of the transmitter and receiver PCB in dB
$ILdd_{PCBmax}(f)$	is the maximum insertion loss of the transmitter and receiver PCB in dB
f	is the frequency in GHz

$$ILdd_{PCB}(f) \geq ILdd_{PCBmin}(f) = 0.0059 + 0.0361\sqrt{f} + 0.0794f \quad (162A-2)$$

for 0.01 GHz $\leq f \leq$ 50 GHz

where

$ILdd_{PCB}(f)$ is the insertion loss of the transmitter and receiver PCB in dB
 $ILdd_{PCBmin}(f)$ is the minimum insertion loss of the transmitter and receiver PCB in dB
 f is the frequency in GHz

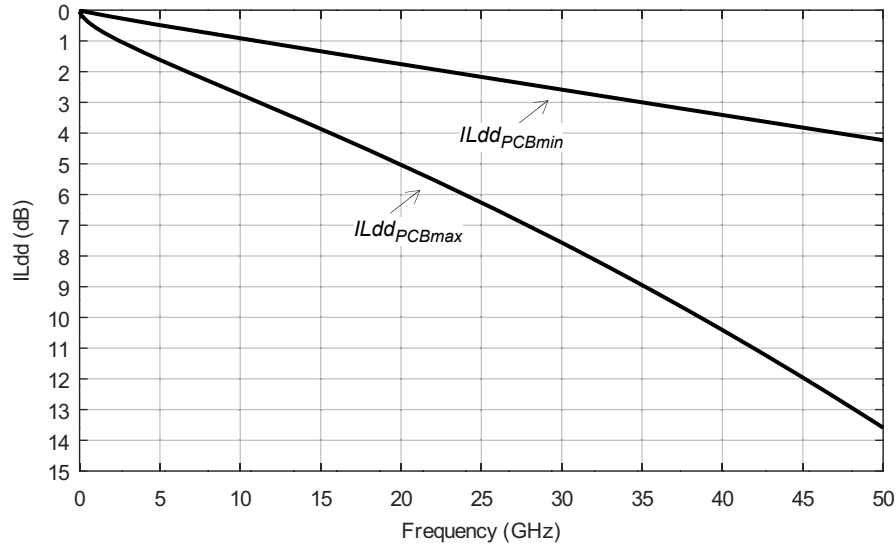


Figure 162A-1—Maximum and minimum insertion loss for transmitter or receiver PCB

The recommended maximum insertion loss from TP0 to TP2 or from TP3 to TP5 including the test fixture is determined using Equation (162A-3), and illustrated in Figure 162A-2. The recommended maximum insertion loss allocation for the transmitter or receiver differential controlled impedance PCBs is consistent with the insertion loss from TP0 to TP2 or TP3 to TP5 and an assumed mated connector insertion loss of 1.6 dB. Note that the recommended maximum insertion loss from TP0 to TP2 or from TP3 to TP5 is 10.975 dB at 26.56 GHz.

$$ILdd_{Host}(f) \leq ILdd_{HostMax}(f) = 1.5658(0.471\sqrt{f} + 0.1194f + 0.002f^2) \quad (162A-3)$$

for $0.01 \text{ GHz} \leq f \leq 50 \text{ GHz}$

where

$ILdd_{Host}(f)$ is the insertion loss from TP0 to TP2 or from TP3 to TP5 including the test fixture in dB

$ILdd_{HostMax}(f)$ is the recommended maximum insertion loss from TP0 to TP2 or from TP3 to TP5 including the test fixture in dB

f is the frequency in GHz

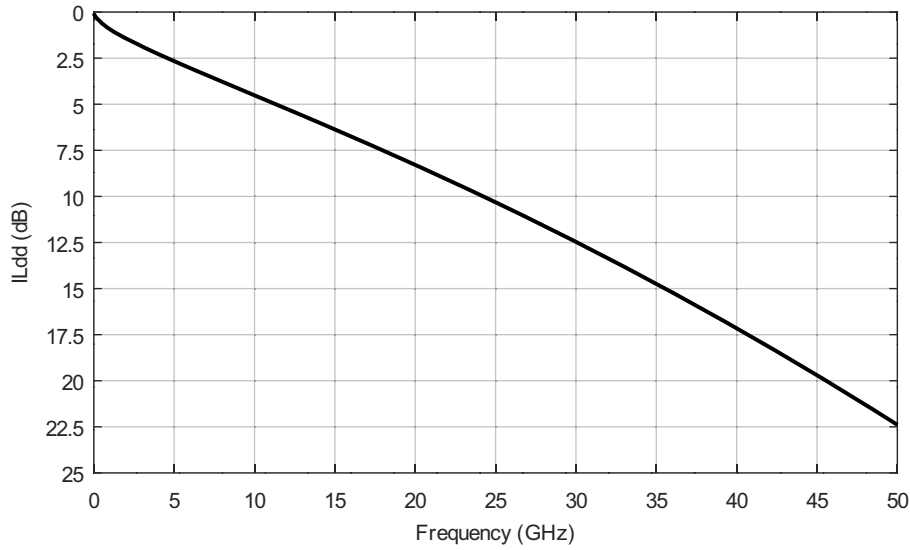


Figure 162A-2—Insertion loss from TP0 to TP2 or from TP3 to TP5

162A.5 Channel insertion loss

This subclause provides information on channel insertion losses for the maximum-loss and minimum-loss cabling topologies.

The maximum channel insertion loss associated with the maximum loss cable assembly is determined using Equation (162A-4). The channel insertion loss budget at 26.56 GHz is illustrated in Figure 162A-3 and Figure 162A-4. Loss budget parameters for all cable topologies are provided in Table 162A-1.

$$ILdd_{Chmax}(f) = ILdd_{Camax}(f) + 2ILdd_{MaxHost}(f) - 2ILdd_{MTFref}(f) \quad (162A-4)$$

for $0.05 \leq f \leq 40$

where

$ILdd_{Chmax}(f)$ is the maximum channel insertion loss in dB between TP0 and TP5
 $ILdd_{Camax}(f)$ is the maximum cable assembly insertion loss in dB (TP1 to TP4)
 $ILdd_{MaxHost}(f)$ is the maximum insertion loss in dB from TP0 to TP2 or TP3 to TP5 using Equation (162A-3)
 $ILdd_{MTFref}(f)$ is the reference insertion loss in dB of the mated test fixture using Equation (162B-5)
 f is the frequency in GHz

The channel insertion loss associated with the minimum insertion loss cable assembly and a maximum insertion loss host channel is determined by Equation (162A-5).

$$ILdd_{Chmin}(f) = ILdd_{Camin}(f) + 2ILdd_{MaxHost}(f) - 2ILdd_{MTFref}(f) \quad (162A-5)$$

for $0.05 \leq f \leq 40$

where

$ILdd_{Chmin}(f)$ is the channel insertion loss in dB between TP0 and TP5 representative of a minimum insertion loss cable assembly and a maximum loss host channel
 $ILdd_{Camin}(f)$ is the minimum cable assembly insertion loss in dB (TP1 to TP4) given in Equation (162-19) and illustrated in Figure 162-8
 $ILdd_{MaxHost}(f)$ is the maximum insertion loss in dB from TP0 to TP2 or TP3 to TP5 using Equation (162A-3)
 $ILdd_{MTFref}(f)$ is the reference insertion loss in dB of the mated test fixture using Equation (162B-5)
 f is the frequency in GHz

Table 162A-1—Insertion loss budget values at 26.56 GHz

Parameter	Value	Units
$ILdd_{Chmax}$	28.5	dB
$ILdd_{Camax}$	19.75	dB
$ILdd_{Chmin}$	19.75	dB
$ILdd_{Camin}$	11.0	dB
$ILdd_{MaxHost}$	10.975	dB
$ILdd_{MTFref}$	6.6	dB

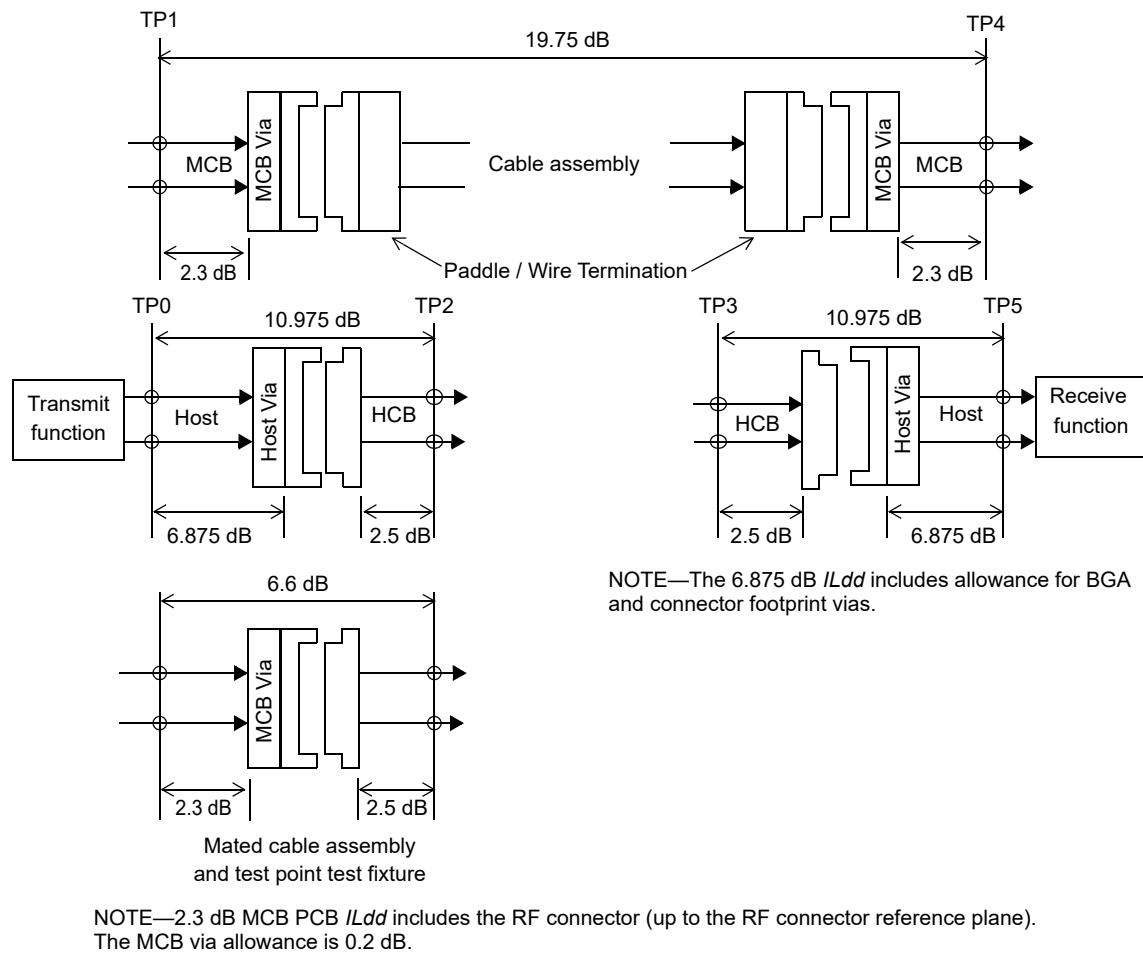
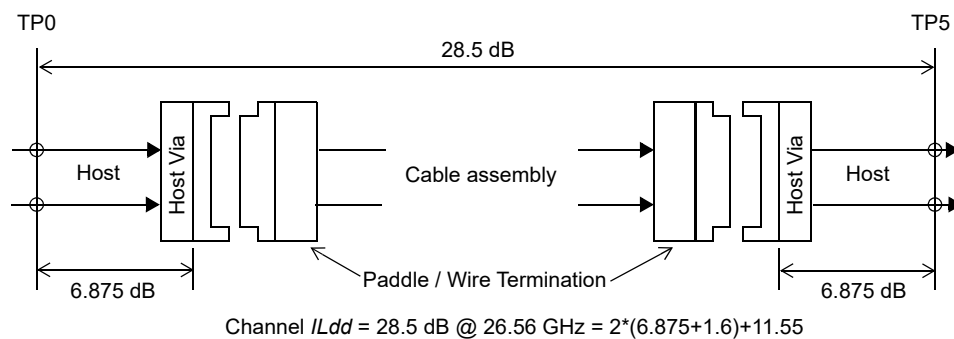


Figure 162A-3—Cable assembly, host, and test fixture insertion loss at 26.56 GHz



NOTE—Channel *ILdd* derived from cable assembly host, and mated test fixture

Figure 162A-4—Channel insertion loss at 26.56 GHz

162A.6 Channel effective return loss (ERL)

The 100GBASE-CR1, 200GBASE-CR2 and 400GBASE-CR4 channels are recommended to meet the ERL specified in 163.10.3.

162A.7 Channel Operating Margin (COM)

The Channel Operating Margin (COM) for the channel between TP0 and TP5, computed using the procedure in 93A.1 and the parameters in Table 163–11, is recommended to be greater than or equal to 3 dB.